An Exponential Succinctness Gap between Three-Variable Logic and the Calculus of Relations

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Abstract

Three-variable first-order logic (FO^3) and the calculus of relations (also known as relation algebra) define the same binary queries, an equivalence that goes back to Tarski's work in the 1940s. The classical translation from FO^3 into the calculus is single-exponential, but it has remained open whether this blowup is necessary. We prove that it is, even over finite structures and even when the translated terms may share subterms. Our proof introduces a new technique: we modify one finite structure in exponentially many small ways that each change the answer of a short FO^3 formula. In any equivalent term, only compositions of binary relations can notice these modifications, and each composition can notice at most two. The same technique gives an exponential succinctness gap between FO^3 and FO^2 sentences, and exponential lower bounds for matrix query languages, where the analogous question was also open.

1 Introduction

Three-variable first-order logic restricted to binary relations (**bin-FO^3**, or simply **FO^3** in this paper) is already a singular point among the finite-variable fragments FO^k of first-order logic. Unlike FO^2, it expresses the existence of a triangle, $\exists x\,\exists y\,\exists z\,(E(x,y)\wedge E(y,z)\wedge E(z,x))$, and transitivity, $\forall x\,\forall y\,\forall z\,(E(x,y)\wedge E(y,z)\to E(x,z))$. Over words, FO^3 expresses all of first-order logic [IK89], and three variables even suffice for all of mathematics: Tarski formalized set theory in $\mathcal{L}^\times$, a variable-free logic of equations between terms of the calculus of relations (**CoR**), and these equations have exactly the expressive power of three-variable sentences [TG87, Giv06]. This expressive strength has sharp computational consequences. For FO^3, even over a single binary relation, satisfiability and finite satisfiability are undecidable, and sentences can force infinite models [KMW62, Ryb22]; for FO^2, both are decidable and finite models suffice [Mor75]. In property testing, a single $\forall\exists\forall$ sentence defines a property of directed graphs that is not testable with a constant number of queries [JZ16]. Symmetric model counting is $\#\mathrm{P}_1$-complete for a fixed FO^3 sentence, whereas it is polynomial for every FO^2 sentence [BVGS15]. Asser's question, open since 1955, whether first-order spectra are closed under complement [Ass55] is equivalent to NE = coNE [JS74] and reduces to spectra of FO^3 sentences [KT15, KT18].

Model checking is solvable in polynomial time for every FO^k, yet FO^3 is singular here as well. A k-variable formula φ can be evaluated naively in time $O(|\varphi|\,n^k)$ on structures with n elements [Var95]. For $k\geq 4$, under the hyperclique hypothesis, this cannot be improved to $O(n^{k-\varepsilon})$, even for a fixed sentence over a single binary relation [NU27, Corollary 32]. For FO^3, however, every sentence φ can be evaluated in time $2^{O(|\varphi|)}n^\omega$ [NU27, Theorem 10], where $\omega<2.3714$ is the exponent of matrix multiplication [ADV+25]. FO^3 and CoR define the same binary queries [TG87], and the key is the central operation of CoR, composition: it eliminates the third variable, $\exists z\,(P(x,z)\wedge Q(z,y))\Leftrightarrow (P;Q)(x,y)$, and it is a Boolean matrix product, computable in subcubic time $O(n^\omega)$. The factor $2^{O(|\varphi|)}$ comes from the translation into CoR. Whether it is necessary sounds like a modern question about query compilation, but it is as old as Tarski's translation from the 1940s [TG87], and its roots reach back to the nineteenth-century calculus of relations (§1.5). It remained open [FGL+15, Remark 4.5], [Nak22, §8], [NU27, Question 3], although under the 3-OV hypothesis (§4.3) no translation can be computed in polynomial time [NU27, Corollary 40]. We settle it: Tarski's translation is optimal in the worst case. Writing T_φ for a smallest CoR term equivalent to φ on finite structures,

$$\varphi \Longrightarrow T_\varphi \qquad |T_\varphi| = 2^{O(|\varphi|)}\ \text{(classical)}, \qquad |T_\varphi| = 2^{\Omega(|\varphi|)}\ \textbf{(our result)}.$$

To state the result formally, we fix the operations of CoR. On a structure with nonempty domain $\mathbb{D}$, CoR uses the constants $\mathbf{0}\coloneqq\varnothing$, $\mathbf{1}\coloneqq\mathbb{D}^2$, and $\mathbf{id}\coloneqq\{(u,u):u\in\mathbb{D}\}$, together with union, intersection, complement $\overline{X}\coloneqq\mathbb{D}^2\setminus X$, converse $X^\smile\coloneqq\{(v,u):(u,v)\in X\}$, and composition

$$X;Y\coloneqq\{(u,v):\text{some } w\in\mathbb{D} \text{ satisfies } (u,w)\in X,(w,v)\in Y\}.$$

Composition implements the elimination step $\exists z\,(P(x,z)\wedge Q(z,y))\Leftrightarrow(P;Q)(x,y)$; a vertex z with $P(x,z)$ and $Q(z,y)$ is a **witness** for (x,y).

1.1 Main results

Let $[k] \coloneqq \{1, \dots, k\}$. Our lower bounds use the following positive formula, for $k \geq 1$; it has one quantifier:

$$G_k(x, y) \coloneqq \exists z \bigwedge_{i=1}^{k} [(P_i(x, y) \wedge Q_i(x, z)) \vee (Q_i(x, y) \wedge P_i(x, z))].$$

Think of P_i and Q_i as two requirements at coordinate i. At a pair (x^*, y^*), the **read edge**, the formula asks for a vertex z^* such that, at every coordinate i, one of the edges (x^*, y^*) and (x^*, z^*) meets P_i and the other meets Q_i, in either order. We call each pair (x^*, z^*) a **witness edge**, and say that the read edge **accepts** it if the two edges satisfy the body of the formula in this way.

Fixing the order at every coordinate gives a disjunctive normal form (DNF). For $A \subseteq [k]$, let $H_A \coloneqq \bigcap_{i \in A} P_i \cap \bigcap_{i \notin A} Q_i$, where an empty intersection is $\mathbf{1}$, and write $A^c \coloneqq [k] \setminus A$. If a read edge accepts a witness edge, let A collect the coordinates at which the read edge meets P_i and the witness edge meets Q_i; at the other coordinates the order is reversed. Then the read edge lies in H_A and the witness edge in H_{A^c}. Conversely, a read edge in H_A accepts every witness edge in H_{A^c}. The **row projection** $X;\mathbf{1}$ holds at (x, y) exactly when x has an outgoing edge in X, so on every nonempty structure

$$G_k \equiv \bigcup_{A \subseteq [k]} \left[H_A \cap (H_{A^c};\mathbf{1})\right],$$

where a formula and a relational expression are identified with the binary queries they define. The DNF uses one composition for each A, 2^k in all.

A CoR **term** T is a syntax tree; $|T|$ counts its nodes and $\#_;(T)$ its compositions. A **relational circuit (DAG)** C allows shared subterms; $|C|$ counts distinct nodes and $g_;(C)$ distinct composition gates. Each input relation, constant, and operation node has unit size. Write $T \equiv_{\text{fin}} \varphi$, or $C \equiv_{\text{fin}} \varphi$, for equivalence on all finite structures over the input signature. Sharing the prefix intersections computes all H_A with $O(2^k)$ nodes, so the DNF gives a relational circuit for G_k with $O(2^k)$ nodes; written as a term, it has $O(k2^k)$ nodes.

Theorem 1.1 (Exponential lower bounds for full CoR)**.** *For every $k \geq 1$, every relational circuit $C \equiv_{\text{fin}} G_k$ has at least 2^{k-1} composition gates, and every* CoR *term $T \equiv_{\text{fin}} G_k$ has at least 2^{k-1} compositions.*

Since G_k has $8k$ syntax-tree nodes, it forces $2^{\Omega(k)}$ compositions from a source of size $O(k)$, even when all other operations are free and arbitrarily numerous. The signature contains only binary relations and grows with k. Together with the single-exponential classical translation, Theorem 1.1 shows that a single exponential is both sufficient and necessary for worst-case FO^3-to-CoR compilation, even with sharing in the target. The theorem is unconditional and counts whole-matrix products rather than scalar operations: it holds even for circuits that are correct only on the structures of our proof, on which G_k can be evaluated in linear time (§4.3).

Theorem 1.1 also shows that the DNF is optimal up to constant factors: no relational circuit uses fewer than half as many compositions as the DNF, and the optimal circuit size is $\Theta(2^k)$. For terms, a finer count determines the optimal size up to a constant factor.

Theorem 1.2 (Term size)**.** *For every $k \geq 1$, every* CoR *term $T \equiv_{\text{fin}} G_k$ has at least $2k2^k - 1$ nodes, and G_k is equivalent to a* CoR *term with $(2k + 6)2^k - 5$ nodes.*

The extra factor k for terms has a concrete origin: singling out one of the 2^k sets A requires examining all k coordinates, and a term pays again for input occurrences that a circuit can share (§5.2).

Where the blowup begins. Theorem 1.1 concerns translations that are correct on all finite structures. The next proposition locates the domain sizes on which the bound for G_k already applies. For a domain size n, let $c_n(G_k)$ be the least number of composition gates in a relational circuit that computes G_k on all n-element structures, and put $n_0(N) \coloneqq \lceil 12N^2 \ln(2N) \rceil + 3$; the structures of our proof are complete graphs with N edge colors, and they exist on every domain of at least $n_0(N)$ elements (§1.3.1).

Proposition 1.3 (Domain size)**.** *For all $k, n \geq 1$,*

$$\max\{2^{j-1} : 1 \leq j \leq k,\ n \geq n_0(2^j)\} \ \leq\ c_n(G_k) \ \leq\ \min\{2^k,\, 4kn\},$$

where the maximum of the empty set is 0. In particular, $c_n(G_k) \geq 2^{k-1}$ for $n \geq n_0(2^k) = O(k4^k)$, and $c_n(G_k) < 2^{k-1}$ for $n < 2^k/(8k)$.

The bound 2^k is the composition count of the DNF, and the bound $4kn$ comes from a circuit that finds, for every vertex x, the at most n combinations of input values on the edges (x, z), one bit at a time (§4.4). The blowup therefore begins at domain size $2^{\Theta(k)}$: every circuit with fewer than 2^{k-1} composition gates errs on one of $2^k + 1$ explicit structures with 4^{k+1} elements, whereas some such circuit is correct on all structures with fewer than $2^k/(8k)$ elements (Corollary 4.8). The composition lower bounds of Theorem 1.1 and Proposition 1.3 follow from one lemma about a single composition on a single finite structure (Lemma 1.4 in §1.3), and so do, through its variants for other operations and targets, the lower bounds of §1.4.

1.2 Natural approaches that fail

Before this work, exponential lower bounds on the size of translations from FO^3 into CoR were known only for $\mathbf{\Sigma_1 CoR}$, even with shared subterms [Nak22, Theorems 6.1 and 6.12]. This fragment of CoR severely restricts complement: complement may be applied only to input relations and **id**, and so never stands above a composition. For example, $\overline{R}; \mathbf{1}$ $(\exists z\, \neg R(x, z))$ lies in Σ_1CoR, but $\overline{R; \mathbf{1}}$ $(\forall z\, \neg R(x, z))$ does not. Nakamura proved this bound with a formula-size game for CoR [Nak22, §4] in the style of Adler and Immerman [AI03]. Organized as a fooling-set argument, the proof becomes simpler and easier to extend. We present it from this standpoint, which also shows exactly where complement breaks it.

In communication complexity, consider a function $f\colon A \times B \to \{0, 1\}$. A **1-rectangle** is a set $A' \times B'$ with $A' \subseteq A$ and $B' \subseteq B$ such that $f(a, b) = 1$ for all $a \in A'$ and $b \in B'$. A **fooling set** is a set $F \subseteq f^{-1}(1)$ such that no 1-rectangle contains two of its elements; equivalently, any two distinct inputs $(a, b), (a', b') \in F$ have a **crossed input** with value 0: $f(a, b') = 0$ or $f(a', b) = 0$. Covering F by 1-rectangles thus takes at least $|F|$ of them [Yao79, LS81, KN97, RY20].

On the side of FO^3, we first build, from Nakamura's instances, a fooling set F for his query φ, which has size $O(k)$. His construction provides $t = \Theta(2^k)$ structures $\mathcal{S}_1, \ldots, \mathcal{S}_t$ in a suitable order, each with exactly three vertices u_i, w_i, v_i, and one structure $\mathcal{N}$ with $\Theta(4^k)$ vertices, two of which, u^* and v^*, are distinguished [Nak22, §5]. To state their properties, write $[\![T]\!]_{\mathcal{S}}$ for the relation that

a term or formula T defines on a structure $\mathcal{S}$, with value $[\![T]\!]_{\mathcal{S}}(u,v) \in \{0,1\}$ at a pair (u,v), and let $E \coloneqq \bigcup_R (R \cup R^\smile)$, where R ranges over the input relations, so that $[\![E]\!]_{\mathcal{S}}$ is the adjacency relation of $\mathcal{S}$. An injective map $\iota\colon \mathcal{S} \hookrightarrow \mathcal{S}'$ is an **embedding** if $[\![R]\!]_{\mathcal{S}'}(\iota(u),\iota(v)) = [\![R]\!]_{\mathcal{S}}(u,v)$ for every input relation R. We write $(\mathcal{S}; a_1, \ldots, a_n) \hookrightarrow (\mathcal{S}'; b_1, \ldots, b_n)$ if some embedding maps $a_1, \ldots, a_n$ to $b_1, \ldots, b_n$, respectively. For all i, the structures satisfy:

(i) $[\![\varphi]\!]_{\mathcal{S}_i}(u_i, v_i) = 1$, but $[\![\varphi]\!]_{\mathcal{N}}(u^*, v^*) = 0$;

(ii) $[\![E]\!]_{\mathcal{S}_i}(u_i, v_i) = [\![E]\!]_{\mathcal{N}}(u^*, v^*) = 0$: the two vertices of each pair are not directly connected;

(iii) $(\mathcal{S}_i; u_i) \hookrightarrow (\mathcal{N}; u^*)$ and $(\mathcal{S}_i; v_i) \hookrightarrow (\mathcal{N}; v^*)$;

(iv) for all $i < j$, some vertex c satisfies $(\mathcal{S}_i; u_i, w_i) \hookrightarrow (\mathcal{N}; u^*, c)$ and $(\mathcal{S}_j; w_j, v_j) \hookrightarrow (\mathcal{N}; c, v^*)$.

The set $F \coloneqq \{(u_i, v_i) : 1 \le i \le t\}$ is a fooling set for φ when φ is evaluated at each pair in the appropriate structure. Indeed, $[\![\varphi]\!]_{\mathcal{S}_i}(u_i, v_i) = 1$ by (i). For $i < j$, the crossed input (u_i, v_j) cannot be evaluated directly, since u_i and v_j lie in different structures; the embeddings of (iv) map them to u^* and v^*, and $[\![\varphi]\!]_{\mathcal{N}}(u^*, v^*) = 0$ by (i).

On the side of CoR, we use that Σ_1CoR is **preserved under extensions**: every Σ_1CoR term T and every embedding $\iota\colon \mathcal{S} \hookrightarrow \mathcal{S}'$ satisfy

$$\iota([\![T]\!]_{\mathcal{S}}) \subseteq [\![T]\!]_{\mathcal{S}'}. \tag{$\diamond$}$$

Let T be a Σ_1CoR term defining φ. Pushing converses down [Nak22, Proposition 6.2], we may assume that T is built by union, intersection, and composition from **literals**: input relations, **id**, and constants, possibly under complement and converse. Say that a term U **separates** $\mathcal{S}_i$ from $\mathcal{N}$ if $[\![U]\!]_{\mathcal{S}_i}(u_i, v_i) = 1$ but $[\![U]\!]_{\mathcal{N}}(u^*, v^*) = 0$; by (i), T separates every $\mathcal{S}_i$ from $\mathcal{N}$. Let T_i be a lowest subterm of T that separates $\mathcal{S}_i$ from $\mathcal{N}$. By (ii), no literal separates, and a union or intersection separates only if an operand does, so T_i is a composition $X;Y$ whose operands do not separate $\mathcal{S}_i$ from $\mathcal{N}$. Its witness for (u_i, v_i) is w_i, that is, $(u_i, w_i) \in [\![X]\!]_{\mathcal{S}_i}$ and $(w_i, v_i) \in [\![Y]\!]_{\mathcal{S}_i}$, as can be verified by (iii) and ($\diamond$). Each T_i thus acts as a 1-rectangle containing (u_i, v_i), and as in communication complexity, no two inputs share one: if $T_i = T_j = X;Y$ with $i < j$, then the witnesses w_i and w_j, (iv), and ($\diamond$) give $[\![X]\!]_{\mathcal{N}}(u^*, c) = [\![Y]\!]_{\mathcal{N}}(c, v^*) = 1$, so $[\![X;Y]\!]_{\mathcal{N}}(u^*, v^*) = 1$, and $X;Y$ does not separate $\mathcal{S}_i$ from $\mathcal{N}$. Hence T has at least t compositions.

Allowing complement above compositions breaks this argument completely, so it cannot be used for full CoR: both of its ingredients fail. First, complement reverses ($\diamond$): $\overline{R;\mathbf{1}}$ holds at (u,v) when u has no R-successor, but fails at $(\iota(u),\iota(v))$ once $\iota(u)$ gains one. Second, the term $\overline{(E \cap (E; \overline{\mathbf{id}}));\mathbf{1}}$, with two compositions, separates every $\mathcal{S}_i$ from $\mathcal{N}$. Other lower-bound methods do not apply directly either; see §1.5.

1.3 Technical overview

Our technique: exact predictions on one reference structure. Our structures are edge-colored complete graphs $\mathcal{M} = (V, \chi)$ with $\chi\colon \binom{V}{2} \to 2^{[k]}$, where the vertex set V is the domain. We write $\chi(u,v) = \chi(v,u)$ for the color of the edge $\{u,v\}$, and set $\chi(u,u) \coloneqq \mathbf{id}$, a label of loops that is not a color. The colors determine the input relations: on an edge of color μ, P_i holds exactly when $i \in \mu$ and Q_i exactly when $i \notin \mu$, and no input relation holds on a loop. Thus a read edge of color μ accepts exactly the witness edges of color μ^c.

A structure (V, χ) has the **two-witness property** if for any distinct $u, v \in V$ and any $\mu, \nu \subseteq [k]$, at least two vertices $z \in V \setminus \{u, v\}$ satisfy $\chi(u, z) = \mu$ and $\chi(z, v) = \nu$. For every V with $|V| \geq n_0(2^k)$, such a structure exists (§1.3.1); we fix one, $\mathcal{M}_0 = (V, \chi)$, the **reference structure**, and an arbitrary vertex $\hbar \in V$, the **hub**. Three consequences of the two-witness property recur in the proof:

(M1) every vertex has edges of every color;

(M2) the value at a pair (u, v) of every relation that a circuit computes on $\mathcal{M}_0$ depends only on $\chi(u, v)$; this needs only one witness for each pair of colors (shown in §1.3.2);

(M3) for any distinct $u, v \in V \setminus \{\hbar\}$ and any colors μ, ν, some $z \notin \{u, v, \hbar\}$ satisfies $\chi(u, z) = \mu$ and $\chi(z, v) = \nu$, as at most one of the two witnesses is $\hbar$; the edges $\{u, z\}$ and $\{z, v\}$ avoid $\hbar$, so recoloring the edges at $\hbar$ leaves them unchanged (used in §1.3.4).

For each color λ, the **fold** of λ recolors the edges of color λ at $\hbar$ to λ^c and leaves all other edges unchanged. With the resulting coloring χ_λ, the **folded structure** is $\mathcal{M}_\lambda := (V, \chi_\lambda)$ (Figure 1).

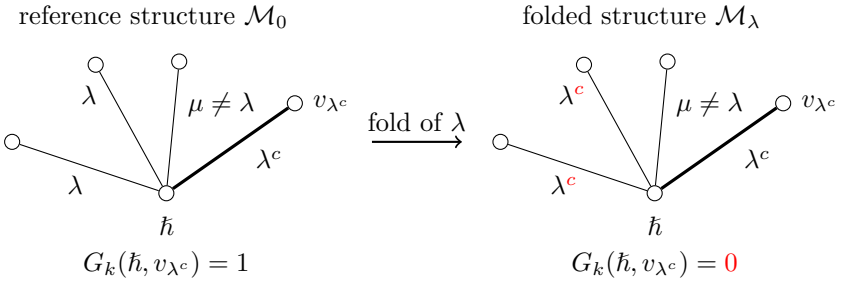


Figure 1: The fold of λ; only edges at the hub $\hbar$ are drawn.

By (M1), we can fix, for each color μ, a vertex v_μ with $\chi(\hbar, v_\mu) = \mu$. The read edge $(\hbar, v_{\lambda^c})$, which the fold of λ does not recolor since $\lambda^c \neq \lambda$, accepts exactly the witness edges of color λ, so $G_k(\hbar, v_{\lambda^c})$ holds exactly when $\hbar$ has an edge of color λ. The hub has such edges in $\mathcal{M}_0$, but not in $\mathcal{M}_\lambda$, where the fold has recolored all of them. Writing $[\![\varphi]\!]_\mathcal{M}$ also for the relation that a gate φ computes on $\mathcal{M}$, we get

$$\mathcal{M}_0 \xrightarrow{\text{fold } \lambda} \mathcal{M}_\lambda \quad (\lambda \subseteq [k]), \qquad [\![G_k]\!]_{\mathcal{M}_0}(\hbar, v_{\lambda^c}) = 1 \neq 0 = [\![G_k]\!]_{\mathcal{M}_\lambda}(\hbar, v_{\lambda^c}).$$

For a color λ, we compare a circuit C on $\mathcal{M}_0$ and on $\mathcal{M}_\lambda$ gate by gate, asking whether a gate g of C satisfies

$$[\![g]\!]_{\mathcal{M}_\lambda}(u, v) = [\![g]\!]_{\mathcal{M}_0}(u', v') \qquad \text{whenever } \chi_\lambda(u, v) = \chi(u', v'). \tag{$\star$}$$

By (M2), the right-hand side depends only on $\chi(u', v')$, so ($\star$) says that g computes on $\mathcal{M}_\lambda$ what its values on $\mathcal{M}_0$ predict from the new colors. Every leaf of C satisfies ($\star$), since both structures obtain their input relations from their colorings by the same rule, and the constants $\mathbf{0}, \mathbf{1}, \mathbf{id}$ depend only on whether a pair is a loop. If C computes G_k, its output gate violates ($\star$) at $(\hbar, v_{\lambda^c})$, which keeps its color while the values above differ.

A gate g **notices** the fold of λ if g, applied on $\mathcal{M}_\lambda$ to the values predicted for its operand gates, does not compute the values predicted for g. The predicted values are determined by values on $\mathcal{M}_0$, so whether g notices the fold of λ depends only on the operation of g and the values of its operand gates on $\mathcal{M}_0$ (§1.3.3). For every gate g, let

$$\Delta_g := \{\lambda : g \text{ violates } (\star) \text{ for } \lambda\}, \qquad D_g := \{\lambda : g \text{ notices the fold of } \lambda\}.$$

The set Δ_g depends on what the circuit below g computes on $\mathcal{M}_\lambda$, whereas D_g depends only on values on $\mathcal{M}_0$. If g has operand gates g_1 and g_2 (for complement and converse, only g_1), a violation at g is inherited from one of them or noticed by g itself: $\Delta_g \subseteq \Delta_{g_1} \cup \Delta_{g_2} \cup D_g$. In a circuit computing G_k, Δ goes from $\varnothing$ at the leaves to $2^{[k]}$ at the output gate,

$$\varnothing = \Delta_{\text{leaf}} \rightsquigarrow \cdots \rightsquigarrow \Delta_{\text{output}} = 2^{[k]},$$

and by the inclusion, a color λ can enter Δ only at a gate that notices the fold of λ, so every color lies in D_g for some gate g (formally, see Lemma 3.3). Unlike the inclusion ($\diamond$) of §1.2, the equality ($\star$) survives complement, which acts pair by pair: Boolean operations and converse notice nothing (§1.3.3), so only compositions notice folds. Using the two-witness property, we show that each composition notices at most two folds (Lemma 1.4 in §1.3.4). Hence a circuit computing G_k needs at least $2^k/2 = 2^{k-1}$ composition gates to notice all 2^k folds; since D_g is attached to the gate, not to its uses, this holds even with sharing.

1.3.1 The reference structure and the tests for G_k

A uniformly random coloring of the edges with N colors has the two-witness property with positive probability whenever $|V| \geq n_0(N)$ (Lemma 3.1), and an explicit construction gives such a coloring for particular values $|V| < 16N^2$ (Appendix A); both go back to Jipsen, Maddux, and Tuza [JMT95]. With the $N = 2^k$ subsets of $[k]$ as colors, such a coloring gives the reference structure. For each color λ, we call $\mathcal{M}_0$, $\mathcal{M}_\lambda$, and the pair $(\hbar, v_{\lambda^c})$ the **test** for λ.

Apart from the tests, the argument uses the rule for the input relations only through the fact that each input relation contains a pair exactly when the label of the pair lies in a fixed set, the same for $\mathcal{M}_0$ and every $\mathcal{M}_\lambda$. The argument therefore works for any **palette**, that is, any set of subsets of $[k]$ used as the colors, provided that the palette is closed under complement, since the fold of λ and the read edge $(\hbar, v_{\lambda^c})$ use the color λ^c; for G_k, there is one test per color. Proposition 1.3 uses palettes of 2^j colors to obtain lower bounds on smaller domains, and §1.4 uses other palettes and input relations for other queries.

1.3.2 Values on the reference structure are color tables

We show (M2): every relation X computed on $\mathcal{M}_0$, viewed as a function from pairs of vertices to $\{0,1\}$, has a **color table**, that is, $X(u,v)$ depends only on the label $\chi(u,v)$. We write $X[\mu]$ for its value at the pairs of color μ and $X[\mathbf{id}]$ for its value on the diagonal; the table of X is the list of these $N+1$ bits, and $S_X := \{\mu : X[\mu] = 1\}$ is the set of colors on which X holds. Operations never change the colors; each operation produces a new relation, whose table is determined by the tables of its operands. For $k = 1$, the colors are $\{1\}$ and $\varnothing$, and some tables are

X	$X[\mathbf{id}]$	$X[\{1\}]$	$X[\varnothing]$
P_1	0	1	0
Q_1	0	0	1
$P_1; Q_1$	0	1	1
$P_1; \mathbf{1}$	1	1	1

For example, $P_1; Q_1$ holds on every edge (u,v), through a vertex z with $\chi(u,z) = \{1\}$ and $\chi(z,v) = \varnothing$, but on no loop, which would need an edge $\{u,z\}$ with both colors; and $P_1; \mathbf{1}$ holds everywhere, including on loops, since every vertex has an edge of color $\{1\}$.

Input relations have tables because the colors determine them, and constants have them trivially; Boolean operations preserve tables, and so does converse, because $\chi(u,v) = \chi(v,u)$. The point is composition. A witness z for a pair (u,v) in $X;Y$ is either an **endpoint witness**, $z \in \{u,v\}$, whose contribution depends only on $X[\mathbf{id}]$, $Y[\mathbf{id}]$, and the entries $X[\mu], Y[\mu]$ at the label μ of (u,v), or an **outside witness**, $z \notin \{u,v\}$. For $u \neq v$, the two-witness property realizes every pair of colors on the edges (u,z) and (z,v) of outside witnesses, so an outside witness exists exactly when S_X and S_Y are both nonempty. For $u = v$, the two edges coincide and every color occurs at u by (M1), so an outside witness exists exactly when $S_X \cap S_Y \neq \varnothing$. Hence, for every color μ,

$$(X;Y)[\mu] = (X[\mathbf{id}] \wedge Y[\mu]) \vee (X[\mu] \wedge Y[\mathbf{id}]) \vee (S_X \neq \varnothing \wedge S_Y \neq \varnothing),$$
$$(X;Y)[\mathbf{id}] = (X[\mathbf{id}] \wedge Y[\mathbf{id}]) \vee (S_X \cap S_Y \neq \varnothing),$$

where a condition stands for its truth value (Lemma 3.2). Consequently *every* relation computed on $\mathcal{M}_0$ has a color table, however much complement is used. Relations computed on a folded structure need not have color tables: on our structures H_λ consists of the edges of color λ, and $[\![H_\lambda; \mathbf{1}]\!]_{\mathcal{M}_\lambda}$ fails at every pair $(\hbar, v)$, since $\hbar$ has no edge of color λ, but holds at the other pairs of the same colors. This is why ($\star$) refers to the values on $\mathcal{M}_0$, and why each folded structure is compared with $\mathcal{M}_0$ directly, never with another folded structure.

1.3.3 Predictions and noticing

For a relation X with a color table, its **prediction** is the relation $X^{(\lambda)}$ on $\mathcal{M}_\lambda$ defined by

$$X^{(\lambda)}(u,v) \coloneqq X[\chi_\lambda(u,v)],$$

so that ($\star$) says $[\![g]\!]_{\mathcal{M}_\lambda} = ([\![g]\!]_{\mathcal{M}_0})^{(\lambda)}$. At a pair (u,v) that the fold does not recolor, $X^{(\lambda)}(u,v) = X(u,v)$; at a recolored edge $(\hbar, z)$, whose color changes from λ to λ^c, the prediction is $X[\lambda^c]$ rather than $X(\hbar, z) = X[\lambda]$. Two simpler invariants would fail. The equality $[\![g]\!]_{\mathcal{M}_\lambda} = [\![g]\!]_{\mathcal{M}_0}$ already fails at the leaves, which differ on the recolored edges. Equality outside the recolored edges cannot be carried through an induction, because a composition moves values from those edges to other pairs.

In this notation, a gate with operation op and operand values $\mathbf{U}$ on $\mathcal{M}_0$ notices the fold of λ exactly when $\mathrm{op}(\mathbf{U}^{(\lambda)}) \neq \mathrm{op}(\mathbf{U})^{(\lambda)}$, where predictions act componentwise on tuples. For a composition gate with operand values X and Y, this reads

$$(X;Y)^{(\lambda)} \neq X^{(\lambda)};Y^{(\lambda)}.$$

Boolean operations, converse, and constants notice nothing, by the identities

$$\overline{X^{(\lambda)}} = \overline{X}^{(\lambda)}, \qquad X^{(\lambda)} \cup Y^{(\lambda)} = (X \cup Y)^{(\lambda)}, \qquad (X^{(\lambda)})^\smile = (X^\smile)^{(\lambda)},$$

and because $\mathbf{0}, \mathbf{1}, \mathbf{id}$ are their own predictions. The general definition also covers the closure and matrix aggregation of §1.4.

1.3.4 Each composition notices at most two folds

On an arbitrary structure, a single composition can react to many separate changes: for $R \coloneqq \{(a, b_i)\}_{i \leq t}$ and $S \coloneqq \{(b_i, c_i)\}_{i \leq t}$ with distinct vertices $b_1, \ldots, b_t, c_1, \ldots, c_t$, removing (a, b_i) from R

removes (a, c_i) from $R; S$, for each i, so the single composition $R; S$ reacts to t separate changes, much as a single term separated all the structures of §1.2. For the folds of the reference structure, this cannot happen. The following bound concerns a single pair of color tables; no circuit is involved.

Lemma 1.4 (Each composition notices at most two folds)**.** *For any two relations X, Y on the reference structure that have color tables, in particular any two relations computed on $\mathcal{M}_0$, we have $(X;Y)^{(\lambda)} = X^{(\lambda)}; Y^{(\lambda)}$ for all colors λ except at most two.*

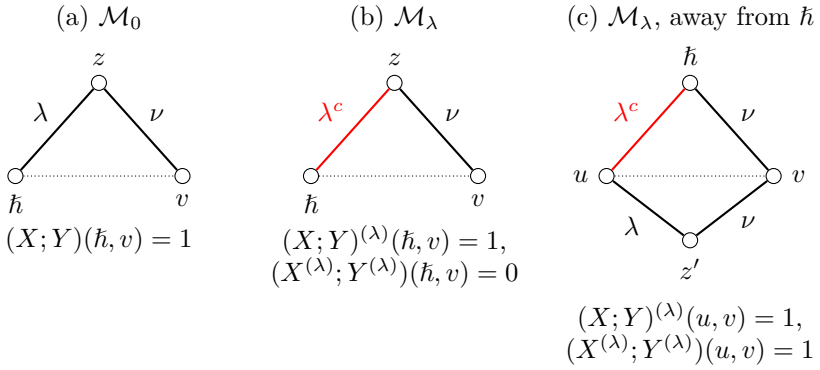


Figure 2: The composition of $X \coloneqq H_\lambda$ and $Y \coloneqq H_\nu$, with $\nu \neq \lambda$, notices the fold of λ at $(\hbar, v)$, but not away from the hub.

Fix X, Y, and λ, and compare the two sides of Lemma 1.4 pair by pair. The left side, the prediction of $X; Y$, is determined by the table of $X; Y$; the right side is what a composition gate computes on $\mathcal{M}_\lambda$ when its operands have the predicted values. Endpoint witnesses contribute the same to both sides, so only outside witnesses matter.

At the hub (Figure 2(a,b)). At a pair $(\hbar, v)$ with $v \neq \hbar$, every outside witness z uses an edge $\{\hbar, z\}$ at the hub, and after the fold these edges carry every color except λ. On the right side, outside witnesses therefore still realize every pair of colors except those whose first color is λ, so the two sides can differ at $(\hbar, v)$ only if λ is the only color on which X holds, that is, $S_X = \{\lambda\}$, as for $X = H_\lambda$ in Figure 2(b). Likewise, they can differ at $(u, \hbar)$ only if $S_Y = \{\lambda\}$, and at $(\hbar, \hbar)$, where a witness uses the single edge $\{\hbar, z\}$ twice, only if $S_X \cap S_Y = \{\lambda\}$.

Away from the hub (Figure 2(c)). At a pair (u, v) with $u \neq v$ and $u, v \neq \hbar$, the fold may cost the witness $\hbar$, but (M3) still provides a witness $z' \notin \{u, v, \hbar\}$ for every pair of colors, so the two sides agree. At (u, u) with $u \neq \hbar$, every color still occurs on an edge at u that avoids $\hbar$, by the two-witness property for the pair $(u, \hbar)$, so the two sides again agree.

Each of the three conditions $S_X = \{\lambda\}$, $S_Y = \{\lambda\}$, and $S_X \cap S_Y = \{\lambda\}$ determines λ. If $S_X \cap S_Y \neq \varnothing$, each of them forces $S_X \cap S_Y = \{\lambda\}$, so at most one fold is noticed; otherwise the third never holds. Hence at most two folds are noticed, and at most one if an operand holds on no color or on every color, as in a row projection $X; \mathbf{1}$ (Lemma 3.4).

1.4 Further results

All lower bounds in Table 1 follow the scheme of §1.3; the entries marked exact are matched by the DNF and by Proposition C.1. Only two ingredients change. The bound two of Lemma 1.4 is replaced by the number of folds that an operation combining values over a witness can notice, while every other operation notices none; and a query yields its tests as in §1.3. The term-size bounds add the occurrence count of §5.2.

Sentences and equations. Our bound also applies to Tarski's correspondence in its equational form, which matches three-variable sentences with equations between relation terms; in this form he formalized set theory [TG87]. A sentence has a single value on each structure, and our tests compare structures: $\mathcal{M}_0$ satisfies G_k at every pair of distinct vertices, while each $\mathcal{M}_\lambda$ has a failing pair, so every test changes the value of the sentence

$$G_k^\forall := \forall x\, \forall y\, (x = y \vee G_k(x, y)).$$

An equation $T = S$ holds exactly when $\overline{\mathbf{1}; ((T \cap \overline{S}) \cup (\overline{T} \cap S)); \mathbf{1}}$ is full, which costs two one-sided compositions, so the argument of Theorem 1.1 applies with a loss of one.

Corollary 1.5 (CoR equations)**.** *If a* CoR *equation $T = S$ holds on exactly those finite structures that satisfy $G_k^\forall$, then every relational circuit with gates computing T and S has at least $2^{k-1} - 1$ composition gates.*

The DNF gives the equation $\mathbf{id} \cup \bigcup_{A \subseteq [k]} [H_A \cap (H_{A^c}; \mathbf{1})] = \mathbf{1}$ with 2^k compositions, so this bound is tight up to a factor of two.

Three versus two variables. To our knowledge, the succinctness of three versus two variables had been studied only on linear orders, where three-variable sentences translate into two-variable sentences of polynomial size [GS05]. Exponential gaps were known for more variables: four versus three on linear orders [GS05], and $k + 1$ versus k for every $k \geq 3$ on general structures with higher-arity relations [VS24]. We show that on general finite structures, three variables are exponentially more succinct than two, already for sentences over binary relations. In an FO^2 formula, every subformula defines a binary relation, like a gate of a relational circuit: connectives notice nothing, and a quantifier is a projection, $\exists y\, \psi \equiv \psi; \mathbf{1}$ or $\exists x\, \psi \equiv \mathbf{1}; \psi$, which notices at most one fold because one operand is constant (§1.3.4). Hence an FO^2 sentence equivalent to $G_k^\forall$ needs a separate quantifier for each of the 2^k tests.

Theorem 1.6 (Three-variable versus two-variable sentences)**.** *For each $k \geq 1$, the sentence $G_k^\forall$ has size $O(k)$ and quantifier depth three over $2k$ binary relations. It is FO^2-definable, but every equivalent FO^2 sentence over finite structures has at least 2^k quantifiers and $k2^k$ relation-atom occurrences. Its minimum FO^2 tree size is $\Theta(k2^k)$.*

The DNF of §1.1 defines G_k in FO^2: writing $H_A(x, y)$ for the conjunction defining H_A gives

$$G_k(x, y) \equiv \bigvee_{A \subseteq [k]} [H_A(x, y) \wedge \exists y\, H_{A^c}(x, y)].$$

This expansion has 2^k quantifiers, and substituting it into $G_k^\forall$ gives $2^k + 2$, within two of the lower bound. The first conjunct checks the read edge. In the second, the endpoint y is no longer needed,

Target or setting	Folds noticed per gate	Lower bound for G_k
Relational circuits	composition 2, projection 1	$g_;(C) \geq 2^{k-1}$ (Thm. 1.1)
Without converse	composition 1 (directed input)	$g_;(C) \geq 2^k$, exact (Cor. 4.6)
CoR equations for $G_k^\forall$	as for circuits	$2^{k-1}-1$ compositions (Cor. 1.5)
FO^2 formulas, sentences	quantifier 1	2^k quantifiers (§5, Thm. 1.6)
Matrix languages, breadth β	matrix–matrix 3β, matrix–vector β	$2^k/(3\beta)$ such products (Thm. 1.8)
With closure X^*	closure 1	$2g_;(C) + g_*(C) \geq 2^k$ (Prop. 1.9)
Randomized, error ε	as for circuits	factor $1-2\varepsilon$ (Cor. 1.10)
Random inputs	as for circuits	success at most $\frac{1}{2} + g/2^k + o(1)$, exact (Cor. 1.11)

Table 1: One argument, many targets. The middle column bounds the number of folds that a single operation of each kind can notice; all other operations notice none. Every bound holds at each domain size at which the reference structure exists.

so its name can be reused for the witness; the binding does not extend to the first conjunct. The short FO^3 formula keeps the read edge and the witness edge in scope together, and so can leave the order at each coordinate open without this expansion. For the binary query G_k, the same tests show that the expansion is optimal.

CoR versus two variables. Our method also separates CoR from FO^2, on a structure with fewer witnesses. On the reference structure, a composition notices at most two folds, so the tests there bound CoR and FO^2 alike, up to a factor of two. A projection, however, still notices at most one fold when every vertex has at least two edges of each atomic type, whereas a composition may then notice many. Let T_k be a CoR term of size $O(k)$ that relates x to y when some R-successor z of x satisfies $P_i(z,z) \leftrightarrow P_i(y,y)$ for every $i \in [k]$. If the fold of $A \subseteq [k]$ removes the R-edges from a hub to the vertices v with $\{i : P_i(v,v)\} = A$, the composition with witness z notices all 2^k folds (§5.4).

Proposition 1.7 (CoR versus FO^2)**.** *For every $k \geq 1$, the* CoR *equation* $\mathbf{id} \cup T_k = \mathbf{1}$ *is* FO^2*-definable, but every equivalent* FO^2 *sentence over finite structures has at least 2^k quantifiers. Every* FO^2 *formula defining T_k also has at least 2^k quantifiers, and this bound is attained.*

Together with Theorems 1.1 and 1.6 and Corollary 1.5, Proposition 1.7 separates FO^3, CoR, and FO^2 exponentially, for binary queries and for sentences, in every direction not covered by the linear translations from FO^2 to CoR [Nak22] and from CoR to FO^3.

Arity elimination in matrix query languages. Brijder, Gyssens, and Van den Bussche asked whether eliminating three-attribute intermediate relations in favor of matrix expressions necessarily incurs an exponential increase in size [BGV19, p. 9]. Matrix query languages replace composition by **aggregation**

$$(X \cdot Y)(u,v) \coloneqq \bigvee_w \kappa(X(u,w), Y(w,v)),$$

where $\bigvee$ is the join of an idempotent semiring $(S, \vee, \otimes, 0_S, 1_S)$ and the **kernel** κ is an arbitrary function of two scalars; with $\kappa = \otimes$, aggregation of matrices and vectors is the usual matrix product.

The semiring has **join breadth** at most β if every finite join equals the join of at most β of its summands. We prove an exponential increase over every nontrivial commutative idempotent semiring of finite join breadth β. Write $\mathcal{G}_k$ for G_k with disjunction, conjunction, and existential quantification replaced by $\vee$, $\otimes$, and the join over witnesses. It has a positive ARA(3) expression of size $O(k)$, where ARA(3) denotes relational algebra on relations whose tuples are annotated with semiring values, with at most three attributes per intermediate relation. A typed matrix circuit computes with $n \times n$ matrices, vectors, and scalars over the semiring.

Theorem 1.8 (Exponential arity-elimination overhead)**.** *For every such semiring and $k \geq 1$, any typed matrix circuit computing $\mathcal{G}_k$ on all finite inputs requires at least $2^k/(3\beta)$ matrix–vector, vector–matrix, or matrix–matrix aggregation gates in total. This holds with arbitrary entrywise functions, each applied in the same way at every entry, and arbitrary kernels in the model of §6.*

Circuits have fixed wiring and no vertex-dependent constant arrays; a kernel receives only its two scalar operands. Transpose and diagonal operations are also allowed. For max–min and tropical semirings, $\beta = 1$. In the proof, at each of the three kinds of pairs at the hub in §1.3.4, $(\hbar, v)$, $(u, \hbar)$, and $(\hbar, \hbar)$, the existence of an outside witness becomes a join over the witnesses. Each join equals the join of at most β of its summands, and a fold can change the join only by removing one of these summands, so at most β folds change each join. Thus a matrix–matrix aggregation notices at most 3β folds, and a matrix–vector or vector–matrix aggregation at most β, whatever the kernel. Section 6 gives sharper operation counts, fixed-size and randomized bounds, and the precise model limits.

Reflexive–transitive closure. The reflexive–transitive closure $X^* \coloneqq \bigcup_{m \geq 0} X^m$ of an $n \times n$ Boolean matrix can be computed in the time of one matrix product, $O(n^\omega)$ [FM71, Mun71, Fur70], although it can replace up to $n - 1$ successive compositions and expresses reachability, which is not first-order. It is thus a natural primitive to add when products are the cost, but it does not avoid the blowup.

Proposition 1.9 (Closure does not help)**.** *Let C be a relational circuit that may also use $X \mapsto X^*$, and let $g_*(C)$ count its distinct closure gates. If $C \equiv_{\mathrm{fin}} G_k$, then $2g_;(C) + g_*(C) \geq 2^k$; in particular, at least 2^{k-1} gates of C are compositions or closures. The bound holds at every domain size at which the reference structure exists.*

Closure therefore does not rescue the speed-up of fast matrix multiplication: once $n \geq n_0(2^k)$, the query G_k of size $O(k)$ still needs 2^{k-1} operations that each cost a matrix product.

Fixed domain sizes, bounded error, and random inputs. The reference structure exists separately at each $n \geq n_0(N)$, and the folds keep its domain, so the bounds hold for **size-specific** circuits, chosen separately for each domain size, as in Proposition 1.3. A randomized circuit is an input-independent distribution of circuits. Its error is **pointwise**: for each input and each pair of vertices, the probability of a wrong answer is at most ε.

Corollary 1.10 (Fixed sizes and bounded error)**.** *Fix $k \geq 1$, $n \geq n_0(2^k)$, and $0 \leq \varepsilon < 1/2$. Every input-independent distribution of relational circuits computing G_k on all n-element inputs with pointwise error at most ε satisfies*

$$\mathbb{E}\, g_;(C) \geq (1 - 2\varepsilon)2^{k-1}.$$

In particular, each exact circuit at each such size needs at least 2^{k-1} composition gates, even when chosen specifically for n.

The size threshold is $O(k4^k)$, and by Proposition 1.3 it cannot be lowered below $2^k/(8k)$. Section 4.3 gives sharper operation-specific statements. The factor $1 - 2\varepsilon$ comes from comparing the two structures of a test: a circuit that cannot distinguish them must err on at least one. The same comparison bounds how much better than guessing a circuit can do on random inputs, and this bound is exact.

Corollary 1.11 (Random inputs)**.** *Fix $k \geq 1$ and n. Color the edges of the complete graph on n vertices independently and uniformly with the colors in $2^{[k]}$, determine P_i, Q_i by the colors as in §1.3, and fix a vertex $\hbar$. Choose a color λ uniformly, a vertex v uniformly among those with $\chi(\hbar, v) = \lambda^c$ (any vertex if there is none), and a fair bit; if the bit is one, apply the fold of λ. Then every relational circuit with g composition gates answers $G_k(\hbar, v)$ correctly with probability at most $\frac{1}{2} + g/2^k + \delta$, where $\delta \coloneqq n^3 4^k \exp(-(n-3)/4^k)$, and for each $g \leq 2^{k-1}$ some circuit with g composition gates answers correctly with probability at least $\frac{1}{2} + g/2^k - \delta$.*

1.5 Related work

Relation algebra and its representations. The calculus of relations predates first-order logic as a formal system: as De Morgan's logic of relations and as the logic of relatives of Peirce and Schröder, it was developed in the nineteenth century [DM64, Pei70, Sch95], decades before Hilbert's lectures and Hilbert and Ackermann's textbook isolated first-order logic [ES13, HA28, Moo88]. Its exact relation to first-order logic emerged gradually. Löwenheim published Korselt's observation that the calculus cannot express every first-order sentence [Löw15], Tarski axiomatized it [Tar41], and Tarski and Givant showed that it defines exactly the binary queries expressible with three variable names [TG87]; see also Maddux [Mad06]. Pratt, Maddux, Burris and Legris, and Brady survey this history [Pra92, Mad91, BL21, Bra00].

For a palette of N colors, the color tables of our reference structure form the finite relation algebra $E_{N+1}(1, 2, 3)$. Jipsen, Maddux, and Tuza gave random and affine-plane representations of this algebra [JMT95]. Their affine-plane coloring, which assigns at least two directions to every color, already gives every pair of distinct vertices two witnesses for every pair of colors when the plane has order at least four; Appendix A verifies this. The probabilistic construction in §3.1 supplies such a coloring separately at every sufficiently large domain size, which our size-specific bounds need. We use these representations as reference structures. The additional ingredient for the lower bound is a bound on how many folds a single operation can notice, uniformly over all values on the reference structure; the existence of the finite representations alone does not provide an operation bound.

Succinctness and lower-bound methods. The classical FO^3-to-CoR translation has a single-exponential size bound, whereas the reverse translation from terms is linear; Appendix F gives the size analysis used here. In prior work, an exponential lower bound was established for positive CoR using two-edge witness queries, whose witness is constrained through both edges (x, z) and (z, y), and extended to the existential fragment Σ_1CoR [Nak22, §6]. That section also treats relational circuit size and gives a superpolynomial separation over a single binary relation symbol. Thus sharing was already covered for restricted target languages; the succinctness question for full CoR remained open. Our bounds allow arbitrary complement, converse, and shared intermediate results together.

Theorem E.1 also gives the full-CoR lower bound for the source family of that work. Our main family uses growing binary signatures, so the single-relation result and our main theorem address different restrictions. The conditional evidence of Nakamura and Uezato for full CoR [NU27, Corollary 40] comes from the 3-OV hypothesis, with the bits of the vectors on vertices; Proposition 4.5 uses the standard hypothesis for two sets of vectors, with the bits on edges, and concerns the specific family G_k.

Formula-size games and techniques derived from them give succinctness gaps for temporal, finite-variable, and modal logics [AI03, GS05, FvdHIK13, HV19]; §1.2 shows that, in full CoR, a term with two compositions already distinguishes Nakamura's structures. Ehrenfeucht–Fraïssé, pebble, and multi-structural games [Fra54, Ehr61, Imm82, Lib04, Imm81, FLRV21, CFI+24] capture quantifier depth, number of variables, and number of quantifiers, but G_k needs only one quantifier and three variables, and its DNF nests no compositions: the gap lies in the number of compositions alone. Succinctness gaps between equally expressive logics have also been obtained by automaton arguments, for instance between FO^2 and unary temporal logic [EVW02]. In communication complexity, rank and discrepancy bounds [MS82, KN97] and Karchmer–Wigderson games [KW90, Juk12] are insensitive to complement, but they measure a single matrix, and free Boolean operations defeat such measures: set disjointness has rank 2^k, yet its 1-entries form the complement of a union of k rectangles. Composition under complement alternates existential and universal quantification over witnesses, much as the communication polynomial hierarchy alternates nondeterminism and co-nondeterminism [BFS86]; there, lower bounds for explicit functions are open even against Arthur–Merlin protocols [GPW18]. We know of no transfer between the two settings. Dawar, Grohe, Kreutzer, and Schweikardt showed that the normal forms provided by Gaifman's locality theorem and by the Łoś–Tarski preservation theorem can require non-elementary size [DGKS07]; our result concerns, in the same spirit, the size cost of Tarski's correspondence between FO^3 and CoR. Our proof exploits the finite algebra of color tables to bound how many folds each distinct operation of a circuit can notice, so it covers circuits with shared gates and unrestricted complement. For the variable-count comparison, Grohe and Schweikardt prove a polynomial translation from three-variable to two-variable sentences on pure finite linear orders [GS05, Theorem 4.3]. On linear orders they also show that four variables are exponentially more succinct than three: a fourth variable permits a divide-and-conquer recursion for distances, and three-variable formulas for distance 2^m need size $2^{\Omega(m)}$ [GS05]. A relational circuit achieves such a recursion by reuse, expressing distance 2^m with $m+1$ compositions, so on linear orders circuits are exponentially more succinct than FO^3. Our lower bounds hold for circuits, so the gap they establish is not a matter of reuse. Vinall-Smeeth's variable-saving separations have at least three target variables and use higher-arity relations [VS24, Theorem 2]. Our sentence separation concerns three versus two variables on general finite structures with growing binary signatures. The linear FO^2-to-CoR translation is recorded in prior work [Nak22]; Proposition 1.7 shows an exponential cost in the other direction, even for FO^2-definable equations.

Matrices and query languages. In databases, Codd's relational algebra [Cod70, Cod72] made operations on whole relations the standard way to evaluate logical queries, and the calculus of relations is the binary core of navigational graph query languages [FGL+15, HPV+21, HVGW22]. A three-variable formula translates in linear size into relational algebra with at most three attributes per intermediate relation (§2), whereas CoR keeps every intermediate relation binary: a composition is a join followed by the projection of the shared attribute. Theorem 1.1 thus measures the cost of eliminating the third attribute. Binary intermediate relations are what fast matrix multiplication

needs, and Boolean matrix products are a current tool for evaluating join–project queries [DHK20, Hu24, AHS25] and first-order graph properties [Wil14, GIKW17, BFK19]. Our bounds count such whole-relation operations; classical scalar-gate lower bounds for matrix products concern a different computation model [JS82, Juk12].

MATLANG combines matrix products, transpose, vector and diagonal operations, and entrywise functions [BGVW19, GMR+21]. For semiring-annotated relations [GKT07], Brijder, Gyssens, and Van den Bussche connect binary-output three-attribute relational algebra with matrix expressions [BGV22]. Their earlier manuscript asks whether exponential arity-elimination overhead is necessary [BGV19, p. 9]. The semiring version $\mathcal{G}_k$ of G_k gives such a lower bound over each fixed nontrivial commutative idempotent semiring of finite join breadth, even with arbitrary kernels in the typed circuit model of §6. That section specifies the permitted operations and separately treats positive nonidempotent computations.

1.6 Organization

Section 2 fixes semantics and resource measures; §3 builds the reference structure, proves that unnoticed folds leave a circuit's values as predicted and that each composition notices at most two folds, and states the test–cost theorem; §4 supplies the tests for G_k, proves Theorem 1.1 and the bounds for small domains, random inputs, and closure, and treats the special case F_k of G_k, in which every Q_i is full, and the relation of both queries to orthogonal vectors; §5 transfers the bound to quantifiers and adds the charging argument for terms (Theorem 1.2), the sentence and equation results, and the separation of CoR from FO^2; §6 extends the analysis of noticed folds to matrix languages; and §7 collects the remaining questions. The appendices provide explicit two-witness colorings, the fixed Hansel circuit, term-size arguments and limits of the method, weighted F_k, directed and two-edge witness queries, and the classical single-exponential translation bound.

2 Preliminaries

Structures and formulas. A signature is a finite set of binary relation symbols, without function or individual constant symbols. A structure $\mathcal{M}$ has a nonempty domain $\mathrm{dom}(\mathcal{M})$ and interprets each symbol R by a binary relation $R^{\mathcal{M}}$. Lower bounds require correctness only on finite structures; the explicit translations used for upper bounds hold on all nonempty structures. The logics FO^2 and FO^3 use only the variable names x, y and x, y, z, respectively, allowing repeated binding of names. Equality is available. The free-variable set of φ is $\mathrm{FV}(\varphi)$. If it is contained in $\{x, y\}$, write $[\![\varphi]\!]_{\mathcal{M}} \subseteq \mathrm{dom}(\mathcal{M})^2$ for its assignment relation. In particular, a sentence denotes the full or empty relation. All syntax uses unary negation and quantification and binary conjunction and disjunction; displayed iterated operations are abbreviations. The size $|\varphi|$ counts nodes after abbreviations are expanded. Thus $|F_k| = 4k$ and $|G_k| = 8k$. Each relation-atom occurrence has unit cost, regardless of the spelling of its relation name. This is a syntax-node measure over growing signatures, rather than a bit-length measure for encoded symbol names.

Evaluation and three-attribute intermediates. There is a direct $O(|\varphi|n^3)$ evaluation algorithm for an FO^3 formula on an n-element domain: store the truth table of each subformula over all assignments to x, y, z. Boolean operations act entrywise. For an existential quantifier, first take the disjunction along its coordinate and then repeat the resulting table along that coordinate. Each

step takes $O(n^3)$ time, and universal quantification uses conjunction instead. In particular, G_k can be evaluated in $O(kn^3)$ time.

The same construction gives a linear-size translation into set relational algebra with at most three attributes per intermediate relation, using the unary domain relation as a constant. Lift every subformula to the common schema $\{x, y, z\}$. For a reflexive atom $R(u, u)$, select the diagonal of R and project to the attribute u; equality atoms are obtained by selection from domain products. Rename ordinary binary atoms as needed, and add missing domain coordinates by Cartesian product. Negation takes the difference from the full three-attribute domain relation. Existential quantification projects away its attribute and then restores that attribute by a Cartesian product with the domain. Each logical operation therefore uses a constant number of algebraic operations, and no intermediate has more than three attributes. For a binary assignment query, finally project to $\{x, y\}$, retaining either coordinate as a dummy if it is not free. This comparison concerns the permitted operations and schemas; our lower bounds below concern computation by the specified whole-relation or whole-matrix operations, rather than tuple-by-tuple evaluation time.

Terms and circuits. The CoR operations and constants have the meanings specified in the introduction. A term T is a syntax tree over binary input symbols, $\mathbf{0}, \mathbf{1}, \mathbf{id}$, union, intersection, complement, converse, and composition. Its value in $\mathcal{M}$ is $[\![T]\!]_{\mathcal{M}}$. Its size $|T|$ is its number of nodes, and $\#_{;}(T)$ counts composition occurrences. A relational circuit (DAG) C uses the same operations and one designated output, but may share intermediate results; $[\![g]\!]_{\mathcal{M}}$ denotes the value in $\mathcal{M}$ of a gate g, and $[\![C]\!]_{\mathcal{M}}$ that of the output. Write $|C|$ for its number of distinct nodes, including inputs, and $g_{;}(C)$ for its number of distinct composition gates. We subsequently say simply *relational circuit*. All operations other than composition may occur without any bound. For a tree,

$$|T| \geq 2\#_{;}(T) + 1,$$

since its leaves outnumber its binary nodes by one. For a term T, $\#_{\mathrm{rel}}(T)$ counts occurrences of input-relation symbols. For a logical formula φ, $\#_{\mathrm{rel}}(\varphi)$ counts occurrences of relation atoms, including reflexive atoms $R(x, x)$, and excludes equality atoms. Quantifier occurrences are counted by $\#_{\exists\forall}(\varphi)$; in a formula circuit we instead count distinct quantifier gates, denoted $g_{\exists\forall}(C)$. Its gates carry variable names and have the usual assignment semantics, so a subformula with given free variables can be shared without renaming.

Write $T \equiv_{\mathrm{fin}} \varphi$ when they define the same binary query on every finite structure over the input signature, and similarly for circuits and formulas. Unsubscripted equivalences used in explicit expansions hold on all nonempty structures. A size-specific circuit may depend on $n = |\operatorname{dom}(\mathcal{M})|$, but not on the input interpretations. No ordering, vertex predicates, named vertices, or arbitrary fixed $n \times n$ arrays are supplied for free. The only relational constants are the empty, full, and identity relations.

Randomized targets. A randomized circuit is an input-independent distribution of deterministic circuits over the permitted operations; its description may depend on n. Pointwise error at most ε means that, for every input and each output position, the reported value is correct with probability at least $1 - \varepsilon$. This need not be simultaneous correctness of the entire output relation. All randomized resource bounds below concern expectations over this distribution, whose support may contain incorrect or differently sized circuits. The same convention applies to randomized formulas and scalar computations.

Set and scalar notation. We use $[k] \coloneqq \{1, \dots, k\}$, $2^{[k]}$ for its power set, $\binom{[k]}{j}$ for its j-element subsets, and $B_k \coloneqq \binom{k}{\lfloor k/2 \rfloor}$. For $I \subseteq [k]$, we write $I^c \coloneqq [k] \setminus I$, and likewise for other subsets of $[k]$. Empty relation intersections and unions mean $\mathbf{1}$ and $\mathbf{0}$; empty conjunctions and disjunctions mean true and false. Relation complement is always denoted by an overline and is relative to the full square of the current domain. Truth values are $\mathbb{B} \coloneqq \{0, 1\}$, and brackets $[\mathcal{P}]$ mean the indicator of a statement $\mathcal{P}$. In the construction, V is the vertex set, $\hbar$ its hub, Λ its palette, and $N \coloneqq |\Lambda|$. Colors are subsets of $[k]$; the letter λ denotes a color that is folded, and μ, ν denote arbitrary colors. A relation X with a color table has entries $X[\mu]$ and $X[\mathbf{id}]$, and $X^{(f)}$ and $X^{(\lambda)}$ denote its predictions under a recoloring f and under the fold of λ (§3.2). The letter d is reserved for the bound of §5.2 on formulas that select one color.

3 Folds, predictions, and the test–cost theorem

This section proves the machinery of the lower bounds in the order of the overview. We construct the reference structure (§3.1) and define color tables and predictions (§3.2). We then show that a relational circuit computes the predicted values on every folded structure whose fold none of its compositions notices (§3.3), and that each composition notices at most two folds (§3.4). With the tests of §4.1, this already proves the composition bounds of Theorem 1.1. Section 3.5 states the test–cost theorem, a counting form of the same argument that also covers randomized circuits and the other targets of §1.4; Section 6 supplies its hypotheses for scalar-valued computations.

3.1 The reference structure

Fix a palette Λ of $N \geq 2$ colors and a vertex set V of size n. A coloring $\chi : \binom{V}{2} \to \Lambda$ assigns a color to every unordered pair of distinct vertices. We write $\chi(u, v)$ for $\chi(\{u, v\})$.

Lemma 3.1 (Two witnesses)**.** *For every integer*

$$n \geq n_0(N) \coloneqq \left\lceil 12N^2 \ln(2N) \right\rceil + 3,$$

there is a coloring such that, for all distinct $u, v \in V$ and all $\mu, \nu \in \Lambda$, at least two distinct vertices $z \in V \setminus \{u, v\}$ satisfy $\chi(u, z) = \mu$ and $\chi(z, v) = \nu$.

Proof. Color the edges independently and uniformly. For fixed distinct u, v, w and colors μ, ν, the probability that no suitable witness occurs outside $\{u, v, w\}$ is at most $\exp(-(n-3)/N^2)$: the candidate witnesses use disjoint pairs of edges. A union bound gives failure probability at most

$$n^3 N^2 \exp(-(n-3)/N^2).$$

At $n = n_0(N)$, the inequality $n_0(N) \leq 13N^3$ bounds this expression by $2197/(4096N) < 1$. It decreases for $n > 3N^2$, proving the assertion at every larger size. Hence some coloring has, for all distinct u, v, w and all colors μ, ν, a witness outside $\{u, v, w\}$; applying this with w equal to one witness for u, v yields a second. □

Fix such a coloring, and choose a vertex $\hbar \in V$, called the hub. The two witnesses of Lemma 3.1 play different roles. One witness outside the endpoints already makes every computed relation

a union of color classes, possibly together with the diagonal (Lemma 3.2). The second ensures that two vertices other than $\hbar$ have, for every pair of colors, a witness that also avoids $\hbar$, so that recoloring the edges at $\hbar$ does not affect pairs away from it (§3.4). Lemma 3.1 also implies that every vertex sees every color, and that every vertex other than $\hbar$ sees every color on edges avoiding $\hbar$ (take $v = \hbar$). For each color μ, fix a vertex v_μ with $\chi(\hbar, v_\mu) = \mu$. No vertex, color, or color neighborhood is supplied as an additional input predicate.

Lemma 3.1 is a two-witness formulation of the colored extension construction of Jipsen, Maddux, and Tuza [JMT95], who gave both random and affine-plane colorings. Their affine-plane construction already supplies two witnesses when the plane has order at least four. We state the property in this form because our proof must keep every color pair available after excluding the hub. Proposition A.1 in Appendix A gives a self-contained two-witness affine-plane construction on fewer than $16N^2$ vertices, and a deterministic construction at every size $n \geq n_0(N)$. The quadratic scale is necessary: for fixed u, v, the N^2 color pairs have disjoint witness sets of size at least two. In the countable homogeneous limit of finite complete graphs with edges colored from Λ [Hod93], every pair of distinct vertices has infinitely many witnesses for every pair of colors; Lemma 3.1 keeps only the finite fragment of this property that our proof uses. On that limit, Lemma 3.2 would follow from homogeneity, and Lemma 3.4, Theorem 3.5, and Lemmas 4.1 and 4.2 would apply unchanged. The limit is pseudo-finite, that is, every first-order sentence true in it has a finite model [Fag76]; each fold is quantifier-free definable in the limit with $\hbar$ named, and naming one element preserves this property. So this route would also give Theorem 1.1, but no explicit domain size such as that of Proposition 1.3.

3.2 Color tables and predictions

For $\mu \in \Lambda$, let $e_\mu \coloneqq \{(u, v) \in V^2 : u \neq v,\ \chi(u, v) = \mu\}$ be the set of edges of color μ, in both orientations. We extend χ to loops by giving every loop the extra label $\mathbf{id}$, which is not a color. A relation $X \subseteq V^2$ has a color table if $X(u, v)$ depends only on $\chi(u, v)$; we then write $X[\mu] \in \mathbb{B}$ for its value at the pairs (u, v) with $\chi(u, v) = \mu$, for $\mu \in \Lambda \cup \{\mathbf{id}\}$. The table of X is the list of these $N + 1$ bits, and

$$S_X \coloneqq \{\mu \in \Lambda : X[\mu] = 1\}$$

is the **off-diagonal support** of X. Thus X is the union of the classes e_μ with $\mu \in S_X$, together with the diagonal if $X[\mathbf{id}] = 1$. A reference structure interprets every input relation by a relation with a color table; we write $\mathcal{M}_0$ for it.

Lemma 3.2 (Closure of color tables)**.** *The constants $\mathbf{0}, \mathbf{1}, \mathbf{id}$ have color tables, and relations with color tables are closed under Boolean operations, converse, and composition. If X and Y have color tables, then for every color $\mu \in \Lambda$,*

$$\begin{aligned} (X;Y)[\mu] &= (X[\mathbf{id}] \wedge Y[\mu]) \vee (X[\mu] \wedge Y[\mathbf{id}]) \vee (S_X \neq \varnothing \wedge S_Y \neq \varnothing), \\ (X;Y)[\mathbf{id}] &= (X[\mathbf{id}] \wedge Y[\mathbf{id}]) \vee (S_X \cap S_Y \neq \varnothing), \end{aligned} \tag{CT}$$

where a condition stands for its truth value.

Proof. Boolean operations act entry by entry, and converse preserves color tables because χ is symmetric. For composition, consider a pair (u, v) and a potential witness z. An endpoint witness, $z \in \{u, v\}$, contributes the disjuncts of (CT) that involve $X[\mathbf{id}]$ or $Y[\mathbf{id}]$. An outside witness,

$z \notin \{u, v\}$, contributes $X[\chi(u, z)] \wedge Y[\chi(z, v)]$. If $u \neq v$, then by Lemma 3.1 the color pairs $(\chi(u, z), \chi(z, v))$ of outside witnesses range over all of $\Lambda \times \Lambda$, so an outside witness exists exactly when S_X and S_Y are both nonempty. If $u = v$, both edges of an outside witness are the edge $\{u, z\}$, and every color occurs on an edge at u, so an outside witness exists exactly when $S_X \cap S_Y \neq \varnothing$. □

In particular, every value of a relational circuit on $\mathcal{M}_0$ has a color table, and

$$e_\mu ; e_\nu = \begin{cases} \mathbf{1}, & \mu = \nu, \\ \overline{\mathbf{id}}, & \mu \neq \nu. \end{cases}$$

Thus the color-table algebra is the symmetric relation algebra $E_{N+1}(1, 2, 3)$ of Jipsen, Maddux, and Tuza [JMT95]; here $\overline{\mathbf{id}} = \bigcup_{\mu \in \Lambda} e_\mu$ is the diversity relation. The proof needs the displayed multiplication table, not automorphisms of the chosen finite graph.

Recolorings and predictions. Given any map $f : \Lambda \to \Lambda$, recolor just the edges at the hub: for $u \neq v$,

$$\chi_f(u, v) \coloneqq \begin{cases} f(\chi(u, v)), & \hbar \in \{u, v\}, \\ \chi(u, v), & \hbar \notin \{u, v\}, \end{cases}$$

and every loop keeps the label **id**. For a relation X with a color table, its prediction $X^{(f)}$ reads the same table on χ_f:

$$X^{(f)}(u, v) \coloneqq X[\chi_f(u, v)] \qquad (u, v \in V).$$

Replacing every input relation by its prediction defines the recolored structure $\mathcal{M}_f$ on the same domain V. Predictions are defined only for relations with color tables, in particular for all values computed on $\mathcal{M}_0$. We do not assume that relations computed on $\mathcal{M}_f$ have color tables, and in general they do not.

Folds. From now on the palette consists of subsets of $[k]$ and is closed under complement $\mu \mapsto \mu^c = [k] \setminus \mu$; since $k \geq 1$, no color equals its complement. The fold of a color λ is the recoloring that sends λ to λ^c and fixes every other color. It recolors every edge of color λ at $\hbar$ to λ^c, so that λ no longer occurs at the hub, and leaves all other edges unchanged. We write χ_λ, $\mathcal{M}_\lambda$, and $X^{(\lambda)}$ for the coloring, the folded structure, and predictions under the fold of λ. Thus $X^{(\lambda)}$ agrees with X except on the recolored edges, where it takes the value $X[\lambda^c]$. Only two properties of the fold matter below: it removes exactly the color λ from the hub, and it changes no edge away from the hub. Any target color other than λ would serve as well.

3.3 Noticing, and unnoticed folds

Let f be a recoloring and $\mathbf{U} = (U_1, \ldots, U_r)$ a tuple of relations with color tables. An r-ary operation op notices f at $\mathbf{U}$ if

$$\mathrm{op}(U_1^{(f)}, \ldots, U_r^{(f)}) \neq \mathrm{op}(\mathbf{U})^{(f)},$$

and it notices the fold of λ if it notices that recoloring. Both sides are determined by $\mathbf{U}$ and f: noticing is a property of the operands on $\mathcal{M}_0$, not of any computation on $\mathcal{M}_f$. Boolean operations, converse, and the constants $\mathbf{0}, \mathbf{1}, \mathbf{id}$ never notice anything. Boolean operations act pair by pair, converse swaps the two orientations of an edge, whose color under χ_f is the same in both orientations,

and the tables of the constants do not depend on the color. Only composition combines values at different pairs. A composition gate of a relational circuit notices the fold of λ if the composition of its operand values on $\mathcal{M}_0$ notices it.

Lemma 3.3 (Unnoticed folds)**.** *Let C be a relational circuit and λ a color. If no composition gate of C notices the fold of λ, then on $\mathcal{M}_\lambda$ every gate of C computes the prediction of its value on $\mathcal{M}_0$. In particular, the outputs of C on $\mathcal{M}_0$ and on $\mathcal{M}_\lambda$ agree at every pair that the fold does not recolor.*

Proof. Evaluate C on $\mathcal{M}_0$. By Lemma 3.2, every gate value has a color table, so its prediction is defined. We show by induction along a topological order of the gates that every gate computes on $\mathcal{M}_\lambda$ the prediction of its value on $\mathcal{M}_0$. For input relations this is the definition of $\mathcal{M}_\lambda$, and constants are their own predictions. Boolean and converse gates preserve the claim because they never notice a fold. If the operands of a composition gate have values X and Y on $\mathcal{M}_0$, then by induction they have values $X^{(\lambda)}$ and $Y^{(\lambda)}$ on $\mathcal{M}_\lambda$. The gate therefore computes $X^{(\lambda)};Y^{(\lambda)}$ on $\mathcal{M}_\lambda$, which equals $(X;Y)^{(\lambda)}$ because the gate does not notice the fold. Finally, every value U on $\mathcal{M}_0$ agrees with $U^{(\lambda)}$ at every pair that the fold does not recolor. □

3.4 Each composition notices at most two folds

Fix X, Y with color tables and a recoloring f, and write $J \coloneqq f(\Lambda)$ for the set of colors at $\hbar$ after the recoloring. Endpoint witnesses contribute the same value to $X^{(f)};Y^{(f)}$ and to $(X;Y)^{(f)}$. At a pair (u,v) with $u \neq v$ and $\mu \coloneqq \chi_f(u,v)$, they contribute $(X[\mathbf{id}] \wedge Y[\mu]) \vee (X[\mu] \wedge Y[\mathbf{id}])$ to $X^{(f)};Y^{(f)}$, because $X^{(f)}$ and $Y^{(f)}$ take the values $X[\mathbf{id}]$ and $Y[\mathbf{id}]$ on the diagonal and $X[\mu]$ and $Y[\mu]$ at (u,v), and the same value to $(X;Y)^{(f)}(u,v) = (X;Y)[\mu]$ by (CT); at a diagonal pair, they contribute $X[\mathbf{id}] \wedge Y[\mathbf{id}]$ to both. The following table compares the outside witnesses. Its left column states when $X^{(f)};Y^{(f)}$ has an outside witness at the pair, and its right column states the last disjunct of (CT), which $(X;Y)^{(f)}$ reads at the pair; here u and v range over vertices other than $\hbar$, and $S \neq \varnothing \neq S'$ means that both sets are nonempty.

Output pair	Outside witness in $X^{(f)};Y^{(f)}$	Last disjunct of (CT)
(u,v), $u \neq v$	$S_X \neq \varnothing \neq S_Y$	$S_X \neq \varnothing \neq S_Y$
(u,u)	$S_X \cap S_Y \neq \varnothing$	$S_X \cap S_Y \neq \varnothing$
$(\hbar,v)$	$S_X \cap J \neq \varnothing \neq S_Y$	$S_X \neq \varnothing \neq S_Y$
$(u,\hbar)$	$S_X \neq \varnothing \neq S_Y \cap J$	$S_X \neq \varnothing \neq S_Y$
$(\hbar,\hbar)$	$S_X \cap S_Y \cap J \neq \varnothing$	$S_X \cap S_Y \neq \varnothing$

For the left column, consider the outside witnesses z of a pair. For distinct $u, v \neq \hbar$, Lemma 3.1 gives, for every pair of colors, an outside witness other than $\hbar$, whose two edges keep their colors; the witness $\hbar$ contributes one more color pair, which is already realized. For (u,u) with $u \neq \hbar$, every color occurs on an edge at u that avoids $\hbar$. At a pair $(\hbar,v)$ of the **hub row**, the left edge $\{\hbar,z\}$ of every outside witness lies at the hub, so its color becomes $f(\chi(\hbar,z))$, while the right edge $\{z,v\}$ keeps its color; by Lemma 3.1 the realized color pairs are exactly $J \times \Lambda$. The pairs $(u,\hbar)$ of the **hub column** are symmetric, with $\Lambda \times J$. At $(\hbar,\hbar)$ both edges of an outside witness are the edge $\{\hbar,z\}$, whose color $f(\chi(\hbar,z))$ ranges exactly over J.

Lemma 3.4 (When a composition notices a recoloring)**.** *Let X and Y have color tables.*

(a) *The composition of X and Y notices the fold of λ only if*

$$S_X = \{\lambda\}, \qquad S_Y = \{\lambda\}, \qquad \text{or} \qquad S_X \cap S_Y = \{\lambda\}.$$

Hence it notices at most two folds, at most one if S_X or S_Y is $\varnothing$ or Λ, and none if both are.

(b) *There is a set $K \subseteq \Lambda$ of at most two colors such that the composition of X and Y does not notice any recoloring f with $K \subseteq f(\Lambda)$. If S_X or S_Y is $\varnothing$ or Λ, then K can be chosen with $|K| \leq 1$, and if both are, with $K = \varnothing$.*

Proof. (a) For the fold of λ, we have $J = \Lambda \setminus \{\lambda\}$. By the table, the two sides can differ only in the hub row, in the hub column, or at $(\hbar, \hbar)$. In the hub row they can differ only if $S_X \neq \varnothing$ but $S_X \cap J = \varnothing$, that is, $S_X = \{\lambda\}$; in the hub column only if $S_Y = \{\lambda\}$; and at $(\hbar, \hbar)$ only if $S_X \cap S_Y = \{\lambda\}$. Each condition determines λ. If $S_X \cap S_Y \neq \varnothing$, each of the first two conditions implies the third, so at most one fold is noticed; otherwise the third condition never holds, and at most two are. If S_Y is $\varnothing$ or Λ, then $S_Y \neq \{\lambda\}$ because $N \geq 2$, and $S_X \cap S_Y$ is $\varnothing$ or S_X, so only the condition $S_X = \{\lambda\}$ remains; the case of S_X is symmetric. If both are $\varnothing$ or Λ, none of the conditions can hold.

(b) If $S_X \cap S_Y \neq \varnothing$, let K consist of one color of $S_X \cap S_Y$. If $K \subseteq J$, then $S_X \cap J$, $S_Y \cap J$, and $S_X \cap S_Y \cap J$ are nonempty, and the two columns of the table agree. Otherwise, let K contain one color of S_X if $S_X \neq \varnothing$, and one color of S_Y if $S_Y \neq \varnothing$. If $K \subseteq J$, then $S_X \cap J$ and $S_Y \cap J$ are nonempty exactly when S_X and S_Y are, and both entries for $(\hbar, \hbar)$ are false. For the sharper assertions, if $S_Y = \varnothing$, all entries are false, and $K = \varnothing$ works. If $S_Y = \Lambda$, then $S_Y \cap J \neq \varnothing$ and $S_X \cap S_Y \cap J = S_X \cap J$, so one color of S_X, if $S_X \neq \varnothing$, suffices. If both S_X and S_Y are $\varnothing$ or Λ, every recoloring gives equal columns. The case of S_X is symmetric. The image of f need only contain the colors of K; f need not fix them. □

The bound two is attained. The composition $e_\mu; e_{\mu^c}$ has $S_X = \{\mu\}$ and $S_Y = \{\mu^c\}$, and notices the folds of both μ and μ^c. On the reference structure of §4.1, the relation e_μ is the relation H_μ of the DNF. Proposition C.1 computes G_k on the reference structure and all its folded structures with 2^{k-1} such compositions. Part (a) gives the lemma stated in the overview.

Lemma 1.4 (Each composition notices at most two folds)**.** *For any two relations X, Y on the reference structure that have color tables, in particular any two relations computed on $\mathcal{M}_0$, we have $(X;Y)^{(\lambda)} = X^{(\lambda)}; Y^{(\lambda)}$ for all colors λ except at most two.*

Proof. Apply Lemma 3.4(a); relations computed on $\mathcal{M}_0$ have color tables by Lemma 3.2. □

By Lemma 1.4, each composition gate notices at most two folds. What a gate notices depends only on its operand values on $\mathcal{M}_0$, so a shared gate counts once, regardless of its fan-out. Hence at most $2g_;(C)$ folds are noticed by some gate of C, and Lemma 3.3 applies to all other folds. Nothing here depends on the query, on which complement-closed palette is used, or on the interpretation of the input relations. If a query has different answers on $\mathcal{M}_0$ and on $\mathcal{M}_\lambda$ at a pair that the fold of λ does not recolor, for each of t colors λ, then every relational circuit C defining it satisfies $2g_;(C) \geq t$. Section 4 supplies such queries.

3.5 Tests and the test–cost theorem

For randomized circuits and for the other targets of §1.4, we use a counting form of this argument that refers only to a few abstract properties. Choose a set $\mathcal{H} \subseteq \Lambda$ of **test colors** and put $t := |\mathcal{H}|$. A test for a query and a color $\lambda \in \mathcal{H}$ consists of the structures $\mathcal{M}_0$ and $\mathcal{M}_\lambda$ and an **output observation** ρ_λ whose two correct values differ, and for which

$$\rho_\lambda(U^{(\lambda)}) = \rho_\lambda(U) \tag{O}$$

for every reference output value U, as defined in (i) below. For binary queries, select a color $\mu \neq \lambda$ and use $\rho_\lambda(U) := U(\hbar, v_\mu)$. The fold does not recolor the pair $(\hbar, v_\mu)$, so (O) holds. Except for the anchored queries of Appendix E.2, our applications use this observation with $\mu = \lambda^c$. Repeating observations for a single removed color does not increase t.

The counting argument uses only three properties of this construction. We call data satisfying them a **test system**:

(i) Each type of value has a collection of **reference values**, which contains the input values on $\mathcal{M}_0$ and is closed under the permitted operations, and every reference value U has a prediction $U^{(\lambda)}$ for each test.

(ii) Every input value on $\mathcal{M}_\lambda$ is the prediction of the same input value on $\mathcal{M}_0$.

(iii) Every test satisfies (O), and its required answers on $\mathcal{M}_0$ and $\mathcal{M}_\lambda$ differ.

For relational circuits, the reference values are the relations with color tables, and Lemma 3.2 supplies (i). The typed scalar computations in §6 will satisfy the same conditions; predictions carry the type of their argument, which we suppress in the notation. For a gate g of a deterministic circuit C, let op_g be its operation and $\mathbf{U}_g$ the tuple of its operand values on $\mathcal{M}_0$. As in §3.3, g notices the fold of λ if $\mathrm{op}_g(\mathbf{U}_g^{(\lambda)}) \neq \mathrm{op}_g(\mathbf{U}_g)^{(\lambda)}$, where predictions act componentwise on tuples. Define

$$D_g := \{\lambda \in \mathcal{H} : g \text{ notices the fold of } \lambda\},$$
$$\mathrm{cost}(C) := \sum_{g \text{ distinct}} |D_g|.$$

For input nodes put $D_g := \varnothing$; constant gates are nullary operations. The definition of D_g compares the entire output of the gate, and it uses predicted operands computed from $\mathcal{M}_0$, not operands recomputed on $\mathcal{M}_\lambda$.

We call $\mathrm{cost}(C)$ the **cost** of C. It counts the pairs of a distinct gate and a test fold that the gate notices. This integer depends on the reference structure and the tests, which remain fixed, but is defined even for an incorrect circuit.

Theorem 3.5 (Test–cost theorem)**.** *Fix a test system with t tests. A deterministic circuit C correct at both observations of every test satisfies*

$$t \leq \mathrm{cost}(C).$$

If C_ω is chosen from an input-independent distribution of circuits, and each observation in each test is incorrect with probability at most $\varepsilon < 1/2$, then

$$(1 - 2\varepsilon)t \leq \mathbb{E}_\omega \,\mathrm{cost}(C_\omega).$$

Proof. Fix a deterministic circuit, not assumed correct, and put $D_C \coloneqq \bigcup_g D_g$. For $\lambda \in \mathcal{H} \setminus D_C$, no gate notices the fold of λ, so the induction in the proof of Lemma 3.3, with (ii) for the input values, gives

$$[\![C]\!]_{\mathcal{M}_\lambda} = ([\![C]\!]_{\mathcal{M}_0})^{(\lambda)}.$$

The equality concerns the whole output. By the observation condition (O), the two observations are equal, whereas their correct values differ. Hence at least one is wrong. If $\mathrm{err}_{\lambda,0}(C), \mathrm{err}_{\lambda,1}(C)$ are the corresponding zero-one error indicators, then

$$\mathrm{cost}(C) + \sum_{\lambda \in \mathcal{H}} (\mathrm{err}_{\lambda,0}(C) + \mathrm{err}_{\lambda,1}(C)) \geq t,$$

because $|D_C| \leq \mathrm{cost}(C)$. Zero error proves the first assertion. Take expectations using the same random circuit on the two structures of each test. Each error indicator has expectation at most ε, which proves the second assertion. No independence among the tests or their errors is required. □

This is the counting argument for all subsequent recoloring bounds. If $\varepsilon_{\lambda,j}$ bounds the error at observation $j \in \{0,1\}$ of the test for λ, the error-count inequality in the proof gives $\mathbb{E}\,\mathrm{cost}(C_\omega) \geq \sum_\lambda (1 - \varepsilon_{\lambda,0} - \varepsilon_{\lambda,1})$. In particular, let $\bar{\varepsilon}$ be the average error probability under the uniform distribution on the $2t$ structure–observation pairs. Then $\mathbb{E}\,\mathrm{cost}(C_\omega) \geq t(1 - 2\bar{\varepsilon})$. For scalar outputs, an error means failure to return the exact required value. Pointwise correctness suffices; simultaneous correctness of all entries is a stronger assumption.

For ordinary CoR, call a composition gate whose operand values on $\mathcal{M}_0$ are X, Y **one-sided** if S_X or S_Y is $\varnothing$ or Λ. Let $m_1(C)$ count these gates and $m_2(C)$ the other composition gates, counting shared gates once.

Lemma 3.6 (Cost of compositions)**.** *For every set of test colors, every relational circuit C satisfies*

$$\mathrm{cost}(C) \leq m_1(C) + 2m_2(C) \leq 2g_;(C).$$

Proof. By Lemma 3.4(a), a one-sided composition notices at most one fold and any other composition at most two. Other gates notice nothing. Summing over distinct gates proves the inequalities. □

The projections $X;\mathbf{1}$ and $\mathbf{1};X$ are one-sided. If a syntactic count is preferred, classify only these projections as one-sided and put every remaining composition in m_2; Lemma 3.6 still holds.

4 Two source queries

The construction of §3 is independent of the query. We now specify its input color tables. In each case the read edge identifies the witness color which a fold removes.

4.1 Complementary colors

For G_k, let $\Lambda = \mathcal{H}$ be a nonempty set of subsets of $[k]$ closed under complement; Theorem 1.1 uses the whole cube $2^{[k]}$. On a color μ, interpret P_i as 1 precisely when $i \in \mu$, and Q_i as 1 precisely when $i \notin \mu$. Put both values zero on every loop. More explicitly,

$$P_i^{\mathcal{M}_0} \coloneqq \bigcup_{\mu \in \Lambda:\, i \in \mu} e_\mu, \qquad Q_i^{\mathcal{M}_0} \coloneqq \bigcup_{\mu \in \Lambda:\, i \notin \mu} e_\mu.$$

For each $\lambda \in \Lambda$, use the read edge $(\hbar, v_{\lambda^c})$, of color λ^c (§3.1), and the fold of λ (§3.2). Since $\lambda \neq \lambda^c$, the fold does not recolor this read edge.

Lemma 4.1 (The tests for G_k)**.** *For every $\lambda \in \Lambda$,*

$$[\![G_k]\!]_{\mathcal{M}_0}(\hbar, v_{\lambda^c}) = 1, \qquad [\![G_k]\!]_{\mathcal{M}_\lambda}(\hbar, v_{\lambda^c}) = 0.$$

Thus G_k supplies $t = |\Lambda|$ tests, and $t = 2^k$ for the whole cube.

Proof. For non-loop edges with colors μ, ν, the body of G_k holds exactly when $\nu = \mu^c$. A loop, whose P_i, Q_i values are all zero, cannot be a witness edge. Against the read color λ^c, the sole accepted witness color is λ, which lies in Λ. That color occurs at $\hbar$ in $\mathcal{M}_0$ and does not occur there after the fold. □

4.2 Disjunctive clauses

For F_k, where $k \geq 2$, take

$$\Lambda := \binom{[k]}{\lfloor k/2 \rfloor} \cup \binom{[k]}{\lceil k/2 \rceil}, \qquad \mathcal{H} := \binom{[k]}{\lceil k/2 \rceil}, \qquad R_i^{\mathcal{M}_0} := \bigcup_{\mu \in \Lambda:\, i \in \mu} e_\mu.$$

All input loops again have value zero. Use the same read colors and folds as in §4.1. The palette consists of the middle layers of the cube: one layer if k is even, and two if k is odd, in which case only the upper layer supplies tests.

Lemma 4.2 (The tests for F_k)**.** *For every $\lambda \in \mathcal{H}$,*

$$[\![F_k]\!]_{\mathcal{M}_0}(\hbar, v_{\lambda^c}) = 1, \qquad [\![F_k]\!]_{\mathcal{M}_\lambda}(\hbar, v_{\lambda^c}) = 0.$$

Thus F_k supplies $t = B_k = \binom{k}{\lfloor k/2 \rfloor}$ tests.

Proof. A witness edge of color $\mu \in \Lambda$ is accepted precisely when $\lambda^c \cup \mu = [k]$, or $\lambda \subseteq \mu$. All palette sets have size at most $|\lambda|$, so this holds precisely for $\mu = \lambda$. Loop witnesses fail because λ is nonempty. The fold removes exactly this accepted color from the hub row. □

A **singleton selector** is a Boolean formula in the tables of the input relations and constants whose support on the entire palette is $\{\lambda\}$ for some test color $\lambda \in \mathcal{H}$; d is a lower bound on its number of input-relation occurrences. The values in the following table are proved in §5.2. This parameter is needed only for the tree-size refinement.

Query	Palette size N	Tests t	Selector size d
G_k	2^k	2^k	k
F_k, even k	B_k	B_k	$k/2$
F_k, odd k	$2B_k$	B_k	$(k+1)/2$

4.3 Consequences and upper bounds

Theorem 1.1 (Exponential lower bounds for full CoR)**.** *For every $k \geq 1$, every relational circuit $C \equiv_{\text{fin}} G_k$ has at least 2^{k-1} composition gates, and every* CoR *term $T \equiv_{\text{fin}} G_k$ has at least 2^{k-1} compositions.*

Proof. Let $C \equiv_{\text{fin}} G_k$, and take the reference structure with palette $2^{[k]}$ at any size $n \geq n_0(2^k)$. For each color λ, the answers of G_k at $(\hbar, v_{\lambda^c})$ on $\mathcal{M}_0$ and on $\mathcal{M}_\lambda$ differ by Lemma 4.1, and the fold of λ does not recolor this pair. By Lemma 3.3, some composition gate of C notices the fold of λ. By Lemma 1.4, each composition gate notices at most two folds, so $2g_{;}(C) \geq 2^k$. A term is a circuit without sharing, so the same bound holds for its number of compositions. □

Together with the expansion of G_k in §1.1, these lower bounds show that the minimum circuit size for G_k is $\Theta(2^k)$; Theorem 1.2, proved in §5.2, determines the minimum term size. For F_k, the tests of Lemma 4.2 give the following bounds.

Proposition 4.3 (Lower bounds for F_k)**.** *For every $k \geq 2$, every relational circuit $C \equiv_{\text{fin}} F_k$ has at least $\lceil B_k/2 \rceil$ composition gates, and every* CoR *term $T \equiv_{\text{fin}} F_k$ has at least $\lceil B_k/2 \rceil$ compositions and at least $2\lceil k/2 \rceil B_k - 1$ nodes.*

Proof. Apply the proof of Theorem 1.1 to the B_k tests of Lemma 4.2. Section 5.2 supplies the tree-size lower bound. □

Corollary 1.10 (Fixed sizes and bounded error)**.** *Fix $k \geq 1$, $n \geq n_0(2^k)$, and $0 \leq \varepsilon < 1/2$. Every input-independent distribution of relational circuits computing G_k on all n-element inputs with pointwise error at most ε satisfies*

$$\mathbb{E}\, g_{;}(C) \geq (1 - 2\varepsilon)2^{k-1}.$$

In particular, each exact circuit at each such size needs at least 2^{k-1} composition gates, even when chosen specifically for n.

Proof. At the prescribed size n, Lemma 3.1 gives the reference structure, and Lemma 4.1 gives all 2^k tests on that domain. Apply the expected-cost assertion of Theorem 3.5 and $\text{cost}(C) \leq 2g_{;}(C)$. Setting $\varepsilon = 0$ includes every exact size-specific circuit. □

More precisely, the same argument gives $\mathbb{E}(m_1 + 2m_2) \geq (1 - 2\varepsilon)t$ for both queries, with $(N, t) = (2^k, 2^k)$ for G_k and the values in the table for F_k. Each bound holds separately at every $n \geq n_0(N)$.

The DNF of §1.1 gives 2^k compositions for G_k, and shared prefix intersections implement it with $O(2^k)$ total nodes. For $I \subseteq [k]$, let $R_I := \bigcap_{i \in I} R_i$. Distributing the conjunction of F_k in the same way gives, on every nonempty structure,

$$F_k \equiv \bigcup_{I \subseteq [k]} [R_I \cap (R_{I^c}; \mathbf{1})].$$

The summand for I fixes the requirements met by the read edge and asks for a witness edge that meets the others. This DNF uses at most $2^k - 1$ compositions, since the summand for $I = [k]$ needs none. Learning a monotone function for each row improves this count.

Proposition 4.4 (A fixed Hansel circuit)**.** *On all nonempty input structures, F_k has a converse-free relational circuit C with*

$$g_{;}(C) \le B_{k+1}, \qquad |C| = O(2^k).$$

Every composition is a row projection. At most $\lfloor k/2 \rfloor + 1$ composition gates occur on any path.

The monotone identification bound is due to Hansel [Han66]; see Aslanyan and Sahakyan [AS17, Theorem 1] for the chain formulation. Appendix B gives a self-contained fixed-circuit implementation, including its Boolean control gates. Thus

$$\lceil B_k/2 \rceil \le \min_{C \equiv_{\mathrm{fin}} F_k} g_{;}(C) \le B_{k+1} \le 2B_k.$$

The order of the composition count is determined. The corresponding total-size interval remains $\Omega(B_k)$ to $O(2^k)$. The corresponding construction for FO^2 formula circuits has at most B_{k+1} distinct quantifier gates (Appendix B).

For both queries, the exponential lower bound is also predicted by fine-grained complexity, for circuits that can be constructed efficiently. The orthogonal vectors problem asks, for sets $A, B \subseteq \{0,1\}^d$ of N vectors each, whether $a \cdot b = 0$ for some $a \in A$ and $b \in B$. The orthogonal vectors hypothesis [Wil05] states that for every $\varepsilon > 0$ there is $c > 0$ such that no algorithm solves this problem with $d = \lceil c \log N \rceil$ in time $O(N^{2-\varepsilon})$. The 3-OV hypothesis is the corresponding statement, with time $N^{3-\varepsilon}$, for deciding whether three sets of vectors contain a triple with no coordinate at which all three vectors are 1 [GIKW17]; Williams's reduction derives both hypotheses from the strong exponential time hypothesis [Wil05]. For a pair (u, v), let $M(u, v) \coloneqq \{i \in [k] : (u, v) \notin R_i\}$ be the set of requirements that it misses. Then $F_k(x, y)$ holds exactly when $M(x, y) \cap M(x, z) = \varnothing$ for some z.

Proposition 4.5 (Orthogonal vectors)**.** *Assume the orthogonal vectors hypothesis. For every $\varepsilon > 0$ there is $c > 0$ such that, for $k = \lceil c \log n \rceil$, no algorithm that receives an n-element structure, given by the adjacency matrices of its input relations, and outputs the relation G_k runs in time $O(n^{3-\varepsilon})$; the same holds for F_k. Consequently, neither query has relational circuits of size $2^{o(k)}$, equivalent to it on finite structures, that can be constructed from k in time $2^{o(k)}$.*

Proof. For the first assertion we may assume $\varepsilon < 1$. Let c' be the constant that the hypothesis provides for $2\varepsilon/3$, and take an instance with N vectors of dimension $d \coloneqq \lceil c' \log N \rceil$. Let $m \coloneqq \lceil N^{2/3} \rceil$, and split A and B into $p \coloneqq \lceil N/m \rceil \le N^{1/3} + 1$ blocks $A_1, \dots, A_p$ and $B_1, \dots, B_p$ of at most m vectors each. The domain consists of p^2 vertices x_{ij}, one for each pair of blocks, and $2m$ further vertices $u_1, \dots, u_m, w_1, \dots, w_m$, so $n = p^2 + 2m = O(N^{2/3})$. Put $k \coloneqq \lceil c \log n \rceil$ with $c \coloneqq 2c'$, so that $k \ge d + 1$ for all large N. Below, a is the s-th vector of A_i and b the t-th vector of B_j.

For F_k, let

$$M(x_{ij}, u_s) \coloneqq \{l \le d : a_l = 1\} \cup \{d + 1\}, \qquad M(x_{ij}, w_t) \coloneqq \{l \le d : b_l = 1\},$$

so that these pairs meet the requirements $d+2, \dots, k$, and let every other pair, including every loop, miss all k requirements; this determines all R_i. In row x_{ij}, the set $M(x_{ij}, u_s)$ contains $d + 1$, so only the pairs (x_{ij}, w_t) with $t \le |B_j|$ can miss a disjoint set, and they do so exactly when $a \cdot b = 0$. Hence $F_k(x_{ij}, u_s)$ holds if and only if a is orthogonal to some vector of B_j.

For G_k, interpret every Q_i as the full relation and P_i as R_i. Each coordinate of the body of G_k then reduces to $R_i(x, y) \vee R_i(x, z)$, so G_k on this structure defines the same relation as F_k on the structure above.

Every pair of vectors meets in some row, so for either query the instance has an orthogonal pair if and only if the query holds at some pair (x_{ij}, u_s) with $s \leq |A_i|$. Either structure is written in time $O(kn^2) = O(N^{4/3} \log N)$, so an evaluation in time $O(n^{3-\varepsilon})$ would solve the instance in time $O(N^{2-2\varepsilon/3})$, contradicting the hypothesis.

For the second assertion, take $0 < \varepsilon < 3-\omega$, where $\omega < 3$ is the exponent of matrix multiplication, and let c be as in the first assertion. On an n-element structure, a composition gate is a Boolean matrix product, computable in time $n^{\omega+o(1)}$, and every other gate takes time $O(n^2)$. For $k = \lceil c \log n \rceil$, a circuit of size $2^{o(k)} = n^{o(1)}$ constructed in time $n^{o(1)}$ would therefore evaluate the query in time $n^{\omega+o(1)} = O(n^{3-\varepsilon})$, contradicting the first assertion. □

The hypothesis thus excludes only circuits that can be constructed efficiently, and the proof gives no explicit exponent, since the constant of the hypothesis is not explicit. Theorem 1.1 and Proposition 4.3 need neither restriction, and by Proposition 4.4 the composition bound for F_k is within a factor 4 of optimal. Nakamura and Uezato [NU27, Corollary 40] derive a related conditional statement for general translations from the 3-OV hypothesis, with the bits of the vectors on vertices. Placing the bits on edges, as above, turns the quadratic hardness of the hypothesis for two sets of vectors into cubic hardness in the number of vertices. With two sets of vectors on vertices, the corresponding query would need only two variables and would have a linear-size translation into CoR [Nak22].

These bounds also do not rest on the difficulty of evaluating the queries. As the proof of Theorem 1.1 shows, their composition bounds hold even for circuits that are correct only on the reference structure and its folded structures, on which every non-loop edge has a color from the palette. For G_k, a read edge there accepts exactly the witness edges of the complementary color, so a trie of the colors present in each row evaluates G_k on all pairs of these structures in time $O(kn^2)$, linear in the size of the input. For F_k, a read edge accepts exactly the witness edges whose color contains the complement of its own color, and at most $\lceil k/2 \rceil + 1$ palette colors do, so the same trie gives time $O(k^2 n^2)$. Proposition 4.5, in contrast, concerns arbitrary inputs.

Corollary 4.6 (Without converse)**.** *Every converse-free relational circuit defining F_k on finite structures has at least B_k composition gates, for $k \geq 2$. Every such circuit defining G_k has at least 2^k composition gates, for $k \geq 1$. The latter bound is exact.*

Appendix E.1 proves the corollary with a directed reference coloring. For full CoR, Proposition C.1 gives a circuit with 2^{k-1} compositions on the symmetric recoloring family itself. Thus that family attains the coefficient two in the local bound. The optimum for G_k on arbitrary inputs remains between 2^{k-1} and 2^k. Appendix E.2 also treats the two-edge witness queries from earlier succinctness work; the full-CoR lower bounds are stated in Theorem E.1.

For Boolean G_k, the factor $1 - 2\varepsilon$ is sharp for the cost of §3.5. Its DNF has cost at most 2^k, and the theorem gives the reverse inequality. Run that exact circuit with probability $1 - 2\varepsilon$, and output $\mathbf{0}$ or $\mathbf{1}$ with probability ε each. Every entry then has error exactly ε, with expected cost $(1 - 2\varepsilon)2^k$. All its compositions are projections, so $m_2 = 0$ and its expected one-sided composition count is also $(1 - 2\varepsilon)2^k$. The same construction attains the expected quantifier bound in §5, and the one-sided operation bounds for Boolean CoR and MATLANG. These equalities use the pointwise-error and

expected-cost conventions of §2.

4.4 Domain size

On arbitrary inputs, the **profile** $\pi(x,z) \in \{0,1\}^{2k}$ of a pair lists the values of $P_1,\dots,P_k,Q_1,\dots,Q_k$ at (x,z); on the reference structure, the profile of an edge is determined by its color. By the definition of G_k, the read edge (x,y) accepts the witness edge (x,z) exactly when, for every i, the profile $\pi(x,y)$ has P_i and $\pi(x,z)$ has Q_i, or $\pi(x,y)$ has Q_i and $\pi(x,z)$ has P_i. Hence $G_k(x,y)$ depends only on $\pi(x,y)$ and the set $\Pi_x \coloneqq \{\pi(x,z) : z \in V\}$ of profiles in row x, which has at most $|V|$ elements.

Proposition 4.7 (Small domains)**.** *For all $k,n \geq 1$ there is a converse-free relational circuit with $4kn$ composition gates, all of them row projections $X;\mathbf{1}$, and $O(k^2n^2)$ other gates, that computes G_k on every structure with at most n elements.*

Proof. A relation is **row-constant** if its value at (x,y) does not depend on y. Every row projection $S;\mathbf{1}$ is row-constant, and it holds in row x exactly when some pair (x,z) lies in S; Boolean combinations of row-constant relations are row-constant. For $1 \leq b \leq 2k$, let E_b be the input relation giving bit b of the profile, and put $E_b^1 \coloneqq E_b$ and $E_b^0 \coloneqq \overline{E_b}$.

In a structure with at most n elements, call a word $w \in \{0,1\}^\ell$ **live** in row x if some profile in Π_x begins with w. Live words of one length begin distinct profiles, so there are at most n of each length, and the empty word is live because $\pi(x,x) \in \Pi_x$. The circuit stores the live words of length ℓ in n slots: row-constant relations $L_{\ell,r}$ and $W_{\ell,r,b}$, for $r \leq n$ and $b \leq \ell$, state that slot r is occupied in row x and that its word has bit b. Initially only slot 1 is occupied, by the empty word. For each slot r and each $\sigma \in \{0,1\}$, the relation

$$S_{\ell,r,\sigma} = L_{\ell,r} \cap \bigcap_{b \leq \ell} \big((E_b \cap W_{\ell,r,b}) \cup (\overline{E_b} \cap \overline{W_{\ell,r,b}})\big) \cap E^\sigma_{\ell+1}$$

holds at (x,z) exactly when slot r is occupied in row x and $\pi(x,z)$ begins with its word followed by σ. The row projection $S_{\ell,r,\sigma};\mathbf{1}$ therefore tells, in every row at once, whether this extension is live. The at most n live extensions are placed into the slots of length $\ell+1$ in the order of their pairs (r,σ). This uses only Boolean operations on the $2n$ answers and the stored words, $O(kn^2)$ gates per length, since the rank of each live extension is a count of earlier live ones. After length $2k$ the occupied slots hold exactly the profiles in Π_x, and

$$G_k = \bigcup_{r \leq n} \Big(L_{2k,r} \cap \bigcap_{i=1}^{k} \big((P_i \cap W_{2k,r,k+i}) \cup (Q_i \cap W_{2k,r,i})\big)\Big),$$

because the intersection over i says that the read edge (x,y) accepts the witness edges whose profile is in slot r. The circuit makes $2n$ row projections at each of the $2k$ lengths, $4kn$ in all, and uses no converse. □

Proposition 1.3 (Domain size)**.** *For all $k,n \geq 1$,*

$$\max\{2^{j-1} : 1 \leq j \leq k,\ n \geq n_0(2^j)\} \ \leq\ c_n(G_k) \ \leq\ \min\{2^k,\, 4kn\},$$

where the maximum of the empty set is 0. In particular, $c_n(G_k) \geq 2^{k-1}$ for $n \geq n_0(2^k) = O(k4^k)$, and $c_n(G_k) < 2^{k-1}$ for $n < 2^k/(8k)$.

Proof. The upper bound 2^k is the DNF of §1.1, and $4kn$ is Proposition 4.7. For the lower bound, let $1 \le j \le k$ with $n \ge n_0(2^j)$, and let Λ_j consist of the sets $A \subseteq [k]$ that contain all or none of the elements $j, \dots, k$. This set is closed under complement and has 2^j elements. Lemma 3.1 gives a coloring of n vertices with palette Λ_j and the two-witness property, and Lemma 4.1 gives 2^j tests on it. A circuit C that computes G_k on n-element structures is correct on both structures of every test. By Lemma 3.3, some composition gate of C notices each of the 2^j folds, and Lemma 1.4 gives $2g_;(C) \ge 2^j$. Finally, $n_0(2^k) = \lceil 12 \cdot 4^k (k+1) \ln 2 \rceil + 3 = O(k4^k)$, and $4kn < 2^{k-1}$ when $n < 2^k/(8k)$. □

The lower bound also holds at the sizes $n = 4^{j+1}$, on the affine colorings of Appendix A. The same holds for formula circuits of FO^2, with quantifier gates in place of composition gates and 2^j in place of 2^{j-1}: the circuit of Proposition 4.7 translates gate by gate, and a quantifier costs at most one (Lemma 5.1).

Corollary 4.8 (Size of counter-models)**.** *Let $k \ge 1$.*

(i) Every relational circuit with fewer than 2^{k-1} composition gates differs from G_k on one of $2^k + 1$ explicit structures with 4^{k+1} elements. A circuit with $g < 2^{k-1}$ composition gates also differs from G_k on a structure with fewer than $64(g+1)^2$ elements.

(ii) For every n, some relational circuit with $4kn$ composition gates agrees with G_k on every structure with at most n elements. For $n < 2^k/(8k)$, this is fewer than 2^{k-1} composition gates.

Proof. (i) Proposition A.1 gives, for the palette $2^{[k]}$, a two-witness coloring on 4^{k+1} vertices; take it with its 2^k folds of §4.1. If a circuit C has fewer than 2^{k-1} composition gates, then by Lemma 1.4 its composition gates notice fewer than 2^k folds. For a fold of a color λ that none of them notices, Lemma 3.3 shows that C takes the same value at $(\hbar, v_{\lambda^c})$ on $\mathcal{M}_0$ and on $\mathcal{M}_\lambda$, whereas G_k does not (Lemma 4.1). For the second claim, let $j \coloneqq \lceil \log_2(g+1) \rceil + 1$. Then $j \le k$ and $2^{j-1} > g$, and the same argument with the palette Λ_j of the preceding proof and its affine coloring on 4^{j+1} vertices applies; since $2^{j-1} < 2(g+1)$, this coloring has fewer than $64(g+1)^2$ vertices. (ii) is Proposition 4.7. □

4.5 Random inputs

Corollary 1.11 (Random inputs)**.** *Fix $k \ge 1$ and n. Color the edges of the complete graph on n vertices independently and uniformly with the colors in $2^{[k]}$, determine P_i, Q_i by the colors as in §1.3, and fix a vertex $\hbar$. Choose a color λ uniformly, a vertex v uniformly among those with $\chi(\hbar, v) = \lambda^c$ (any vertex if there is none), and a fair bit; if the bit is one, apply the fold of λ. Then every relational circuit with g composition gates answers $G_k(\hbar, v)$ correctly with probability at most $\frac{1}{2} + g/2^k + \delta$, where $\delta \coloneqq n^3 4^k \exp(-(n-3)/4^k)$, and for each $g \le 2^{k-1}$ some circuit with g composition gates answers correctly with probability at least $\frac{1}{2} + g/2^k - \delta$.*

Proof. Let $\mathcal{W}$ be the event that the coloring has the two-witness property. The proof of Lemma 3.1, with $N = 2^k$, gives $\Pr[\neg\mathcal{W}] \le \delta$. Fix a coloring in $\mathcal{W}$. Then $\hbar$ has edges of every color, so v is uniform among the vertices with $\chi(\hbar, v) = \lambda^c$. For each color λ, let $\mathrm{err}_{\lambda,0}, \mathrm{err}_{\lambda,1}$ indicate an error of C at $(\hbar, v_{\lambda^c})$ on $\mathcal{M}_0$ and on $\mathcal{M}_\lambda$. The error-count inequality in the proof of Theorem 3.5, with the

tests of Lemma 4.1 and Lemma 3.6, gives

$$\sum_{\lambda}(\mathrm{err}_{\lambda,0}+\mathrm{err}_{\lambda,1}) \geq 2^k - \mathrm{cost}(C) \geq 2^k - 2g$$

for every choice of the vertices v_μ of §3.1, hence also in expectation when each v_μ is uniform among the vertices with $\chi(\hbar, v_\mu) = \mu$. Given the coloring, the probability of an error is the average of $\mathrm{err}_{\lambda,b}$ over uniform λ and b, so it is at least $(2^k - 2g)/2^{k+1} = \frac{1}{2} - g/2^k$. Averaging over the colorings proves the upper bound.

For the lower bound, choose g of the pairs $\{\lambda, \lambda^c\}$ and keep, in the circuit of Proposition C.1, only the terms for these pairs. This circuit has g composition gates. On a coloring in $\mathcal{W}$, it is correct on both structures of the $2g$ tests whose colors lie in the chosen pairs, and for every other color it answers zero at the read edge, which is correct when the bit is one. Its success probability is therefore at least $(1-\delta)(\frac{1}{2} + g/2^k) \geq \frac{1}{2} + g/2^k - \delta$. □

4.6 Reflexive–transitive closure

Proposition 1.9 (Closure does not help)**.** *Let C be a relational circuit that may also use $X \mapsto X^*$, and let $g_*(C)$ count its distinct closure gates. If $C \equiv_{\mathrm{fin}} G_k$, then $2g_{;}(C) + g_*(C) \geq 2^k$; in particular, at least 2^{k-1} gates of C are compositions or closures. The bound holds at every domain size at which the reference structure exists.*

Proof. On the reference structure, (CT) gives $X; X = \mathbf{1}$ whenever the off-diagonal support S_X is nonempty. Hence $X^* = \mathbf{1}$ if $S_X \neq \varnothing$ and $X^* = \mathbf{id}$ otherwise; in particular, closure preserves color tables, as condition (i) of a test system requires. Let λ be a test color, and put $J \coloneqq \Lambda \setminus \{\lambda\}$. If $S_X = \varnothing$, then $X^{(\lambda)} = X$, and both $(X^{(\lambda)})^*$ and $(X^*)^{(\lambda)}$ equal $\mathbf{id}$. If S_X contains a color in J, the table of §3.4 gives $X^{(\lambda)}; X^{(\lambda)} = \mathbf{1}$, so $(X^{(\lambda)})^* = \mathbf{1} = (X^*)^{(\lambda)}$. Thus a closure gate notices the fold of λ only when $S_X = \{\lambda\}$, so it notices at most one fold. Theorem 3.5 and Lemma 3.6 give $2^k \leq m_1(C) + 2m_2(C) + g_*(C) \leq 2g_{;}(C) + g_*(C)$ with the tests of Lemma 4.1. □

With the tests of Lemma 4.2, the same argument gives $2g_{;}(C) + g_*(C) \geq B_k$ for every circuit with closure that defines F_k.

5 Two-variable logic and tree size

The formulas F_k, G_k are FO^2-definable, but their displayed linear-size forms use three variable names. This section quantifies the difference. Section 5.4 also separates CoR from FO^2.

5.1 A translation that preserves the relevant resources

For a formula φ using only x, y, view its value as a binary relation of assignments, even if one or both variables are absent from $\mathrm{FV}(\varphi)$. Thus a sentence denotes $\mathbf{1}$ or $\mathbf{0}$. The standard linear

translation τ into CoR is as follows:

φ	$\tau(\varphi)$	φ	$\tau(\varphi)$
$R(x,y)$	R	$R(y,x)$	$R^\smile$
$R(x,x)$	$(R \cap \mathbf{id}); \mathbf{1}$	$R(y,y)$	$\mathbf{1}; (R \cap \mathbf{id})$
$x = y$ or $y = x$	$\mathbf{id}$	$x = x$ or $y = y$	$\mathbf{1}$
$\exists x\, \psi$	$\mathbf{1}; \tau(\psi)$	$\exists y\, \psi$	$\tau(\psi); \mathbf{1}$

Conjunction, disjunction, and negation translate to intersection, union, and complement. Universal quantifiers use $\forall u\, \psi \equiv \neg\exists u\, \neg\psi$, for $u \in \{x, y\}$. Induction on formulas proves correctness on all nonempty structures. The linear size bound is recorded in [Nak22, Remark 3.22].

Translation preserves the relation-atom count $\#_{\mathrm{rel}}$ exactly. For the reference structures of §§4.1–4.2, the off-diagonal support of $R \cap \mathbf{id}$ is empty. Hence the extra composition used for a reflexive atom notices no fold. Every quantifier contributes a one-sided composition, which notices at most one fold. These observations give the following resource bound.

Lemma 5.1 (Quantifier cost)**.** *For either source-query test system, every* FO^2 *formula* φ *satisfies*

$$\mathrm{cost}(\tau(\varphi)) \le \#_{\exists\forall}(\varphi).$$

For a formula circuit, the cost is at most its number of distinct quantifier gates. The translation preserves relation-atom occurrences exactly for formulas.

This is a statement about cost, not about the syntactic composition count: the latter also counts the compositions introduced by reflexive atoms, which notice no fold.

Combining Lemma 5.1 with Theorem 3.5 gives

$$\varphi \equiv_{\mathrm{fin}} G_k \Longrightarrow \#_{\exists\forall}(\varphi) \ge 2^k, \qquad \varphi \equiv_{\mathrm{fin}} F_k \Longrightarrow \#_{\exists\forall}(\varphi) \ge B_k.$$

The FO^2 expansions obtained from the DNFs of §1.1 and §4.3 use exactly 2^k quantifiers for G_k and at most $2^k - 1$ for F_k. In particular, the optimum quantifier count for G_k is exactly 2^k. At a fixed size $n \ge n_0(N)$ these conclusions still hold. For a distribution of FO^2 formulas or circuits with pointwise error at most ε, the corresponding expected quantifier count is at least $(1 - 2\varepsilon)t$. No restriction to existential target formulas is made.

5.2 Counting input occurrences in trees

The circuit theorem also yields a stronger bound for syntax trees. A singleton selector for a test color $\lambda \in \mathcal{H}$ is a Boolean formula in the tables of the input relations, with constants allowed, whose support on the entire palette is exactly $\{\lambda\}$.

Lemma 5.2 (Disjoint input-occurrence charging)**.** *Fix the reference structure and the test folds of §3.5. Suppose every singleton selector of a test color requires at least* $d > 0$ *input-relation occurrences. Then every* CoR *term* T*, whether correct or not, satisfies*

$$\mathrm{cost}(T) \le \#_{\mathrm{rel}}(T)/d.$$

Appendix C.1 proves the lemma by assigning disjoint sets of input occurrences to composition nodes. The disjointness uses the tree structure of a term.

For the cube used by G_k, omitting coordinate i leaves a color indistinguishable from its i-th-bit flip, even when both P_i, Q_i are available. Hence $d = k$. For F_{2m}, reading fewer than m coordinates leaves an unread element in each of λ and λ^c; swapping them preserves the observed values. For F_{2m+1}, reading fewer than $m+1$ coordinates leaves an unread element of the test color λ; deleting it gives another palette color with the same observed values. Thus $d = \lceil k/2 \rceil$ for F_k. The conjunction of the positive coordinates of λ selects λ in the F_k palette, and H_λ selects λ in the G_k palette, so these selector bounds are exact. Theorem 3.5, Lemma 5.2, and preservation of $\#_{\mathrm{rel}}$ under τ give

	F_k	G_k
$\#_{\mathrm{rel}}(T),\ \#_{\mathrm{rel}}(\varphi)$	$\geq \lceil k/2 \rceil B_k$	$\geq k2^k$

as lower bounds whenever T or $\varphi \in \mathrm{FO}^2$ defines the indicated query. A unary/binary syntax tree with L leaves has at least $2L - 1$ nodes; each quantifier is an additional unary node. Therefore

$$
\begin{aligned}
&|T| \geq 2\lceil k/2 \rceil B_k - 1 && (T \equiv_{\mathrm{fin}} F_k),\\
&|T| \geq 2k2^k - 1 && (T \equiv_{\mathrm{fin}} G_k),\\
&|\varphi| \geq (2\lceil k/2 \rceil + 1)B_k - 1 && (\varphi \in \mathrm{FO}^2,\ \varphi \equiv_{\mathrm{fin}} F_k),\\
&|\varphi| \geq (2k+1)2^k - 1 && (\varphi \in \mathrm{FO}^2,\ \varphi \equiv_{\mathrm{fin}} G_k).
\end{aligned}
$$

The second inequality is the lower bound of the term-size theorem stated in the introduction.

Theorem 1.2 (Term size)**.** *For every $k \geq 1$, every* CoR *term $T \equiv_{\mathrm{fin}} G_k$ has at least $2k2^k - 1$ nodes, and G_k is equivalent to a* CoR *term with $(2k+6)2^k - 5$ nodes.*

Proof. The lower bound is the second inequality above. Factoring the read-edge tests gives a term with $(2k+6)2^k - 5$ nodes that is equivalent to G_k on all nonempty structures (Appendix C.2). □

The same factoring gives an FO^2 tree for G_k with $(2k+5)2^k - 5$ nodes, so both optimal tree sizes for G_k are $\Theta(k2^k)$. For F_k the established tree-size interval is $\Omega(kB_k)$ to $O(k2^k)$. Combining Lemma 5.2 with the randomized test–cost bound yields $\mathbb{E}\#_{\mathrm{rel}}(T) \geq d(1-2\varepsilon)t$ for randomized terms, and the same bound holds for randomized FO^2 formulas.

5.3 Sentences and the variable hierarchy

Recall the sentence $G_k^\forall = \forall x\, \forall y\, (x = y \vee G_k(x, y))$ from the introduction. We now prove the sentence separation stated in the introduction.

Theorem 1.6 (Three-variable versus two-variable sentences)**.** *For each $k \geq 1$, the sentence $G_k^\forall$ has size $O(k)$ and quantifier depth three over $2k$ binary relations. It is FO^2-definable, but every equivalent FO^2 sentence over finite structures has at least 2^k quantifiers and $k2^k$ relation-atom occurrences. Its minimum FO^2 tree size is $\Theta(k2^k)$.*

Proof. The source has size $8k + 4$ and quantifier depth three. It is true in $\mathcal{M}_0$: every off-diagonal read color has a complementary witness color, and equality handles the diagonal. It is false in every $\mathcal{M}_\lambda$, as witnessed by $(\hbar, v_{\lambda^c})$.

A sentence is represented by a constant binary relation, and either constant is its own prediction. Thus these structures give 2^k tests for the sentence itself. For any equivalent FO^2 sentence φ,

apply Theorem 3.5 to $\tau(\varphi)$. Lemma 5.1 gives $\#_{\exists\forall}(\varphi) \geq 2^k$, and Lemma 5.2, together with exact preservation of $\#_{\mathrm{rel}}$, gives $\#_{\mathrm{rel}}(\varphi) \geq k2^k$. Conversely, substitute the FO^2 expansion of G_k into the sentence. It has $2^k + 2$ quantifiers and $O(k2^k)$ nodes. This proves FO^2-definability and the matching order of tree size. □

The quantifier optimum for this sentence is therefore in $[2^k, 2^k + 2]$; the exact 2^k optimum proved earlier concerns the binary query. The lower bounds hold separately at every $n \geq n_0(2^k)$. For randomized FO^2 sentences with error at most ε on each input, the same tests give expected quantifier count at least $(1 - 2\varepsilon)2^k$ and expected relation-atom count at least $(1 - 2\varepsilon)k2^k$.

This complements the variable-saving separations of Vinall-Smeeth [VS24, Theorem 2], whose stated range starts at three variables on the target side. Our result concerns the remaining comparison of three and two variables, over growing binary signatures. It does not claim a lower bound for separating a single prescribed pair of structures as a function of their size. On pure finite linear orders, in contrast, Grohe and Schweikardt [GS05, Theorem 4.3] give a polynomial translation from three-variable sentences to two-variable sentences. The qualifications "pure orders" and "sentences" are essential to that comparison.

Equations. The same tests bound the equational form of Tarski's correspondence.

Corollary 1.5 (CoR equations)**.** *If a* CoR *equation $T = S$ holds on exactly those finite structures that satisfy $G_k^{\forall}$, then every relational circuit with gates computing T and S has at least $2^{k-1} - 1$ composition gates.*

Proof. Let C have gates computing T and S, and add gates for

$$E \coloneqq \overline{\mathbf{1}; \left((T \cap \overline{S}) \cup (\overline{T} \cap S)\right); \mathbf{1}},$$

which is the full relation when $T = S$ and empty otherwise. This adds two compositions, both one-sided. By the proof of Theorem 1.6, $G_k^{\forall}$ is true in $\mathcal{M}_0$ and false in every $\mathcal{M}_\lambda$, so E is full on $\mathcal{M}_0$ and empty on every $\mathcal{M}_\lambda$. Both constants are their own predictions, so these are 2^k tests, and Theorem 3.5 with Lemma 3.6 gives $2^k \leq 2g_{;}(C) + 2$. □

5.4 Separating CoR from FO^2

The linear translation above goes from FO^2 to CoR. On the reference structure, a composition notices at most two folds and a quantifier at most one, so tests on the reference structure bound CoR circuits and FO^2 formulas alike, up to a factor of two. To separate the two formalisms, we use a structure on which a quantifier still notices at most one fold, but a single composition notices all of them.

Use input symbols $R, P_1, \ldots, P_k$, and put

$$D_i^1 \coloneqq \mathbf{id} \cap P_i, \qquad D_i^0 \coloneqq \mathbf{id} \cap \overline{P_i}, \qquad T_k \coloneqq R; \bigcap_{i=1}^{k} \left[(D_i^0; \mathbf{1}; D_i^0) \cup (D_i^1; \mathbf{1}; D_i^1) \right].$$

The complement in D_i^0 is taken before intersection with the identity; it is not the complement of D_i^1. Expanding these abbreviations gives an $O(k)$-size tree. For a vertex v, call $\pi(v) \coloneqq \{i \in [k] : P_i(v, v)\}$

its **loop pattern**. Then $T_k(x,y)$ holds exactly when some R-successor z of x satisfies $\pi(z) = \pi(y)$, that is, when

$$\varphi_k(x,y) \coloneqq \exists z \left(R(x,z) \wedge \bigwedge_{i=1}^{k} (P_i(z,z) \leftrightarrow P_i(y,y)) \right)$$

holds. Here $\alpha \leftrightarrow \gamma$ abbreviates $(\alpha \wedge \gamma) \vee (\neg\alpha \wedge \neg\gamma)$.

The structures. Let $V \coloneqq 2^{[k]} \times \mathbb{Z}_{11}$ and $J \coloneqq \{1,2,3,4,7,8\} \subseteq \mathbb{Z}_{11}$. The structure $\mathcal{W}_0$ on V has the loops $P_i((A,s),(A,s))$ for $i \in A$, no other P_i-entries, and $R((A,s),(B,t))$ exactly when $t - s \in J$; thus $\pi(A,s) = A$, and R has no loops. The **atomic type** of a pair (u,v) of distinct vertices is $(R(u,v), R(v,u), \pi(u), \pi(v))$, and that of a loop (u,u) is $\pi(u)$; it determines the values of all atomic formulas at the pair. As $d \coloneqq t - s$ ranges over $\mathbb{Z}_{11} \setminus \{0\}$, the truth values of $d \in J$ and $-d \in J$ take each of the four combinations at least twice: $(1,0)$ at $d \in \{1,2\}$, $(0,1)$ at $d \in \{9,10\}$, $(1,1)$ at $d \in \{3,4,7,8\}$, and $(0,0)$ at $d \in \{5,6\}$. Hence every vertex u is the first entry of at least two pairs of each atomic type $(r, r', \pi(u), B)$, and the second entry of at least two pairs of each atomic type $(r, r', B, \pi(u))$.

Fix the hub $\hbar \coloneqq (\varnothing, 0)$. For $A \subseteq [k]$, the fold of A yields the structure $\mathcal{W}_A$ obtained from $\mathcal{W}_0$ by removing from R the edges $(\hbar, v)$ with $v \neq \hbar$ and $\pi(v) = A$. It changes the atomic types of these pairs and of their converses, and of no other pair. Afterwards, no pair with first entry $\hbar$ has a type $(1, r', \varnothing, A)$, and no pair with second entry $\hbar$ has a type $(r, 1, A, \varnothing)$; all other types still occur at $\hbar$.

Type tables and predictions. A relation $X \subseteq V^2$ has a **type table** if $X(u,v)$ depends only on the atomic type of (u,v) in $\mathcal{W}_0$; let S_X be the set of atomic types at which X holds. Every atomic type of $\mathcal{W}_A$ occurs in $\mathcal{W}_0$, so the prediction $X^{(A)}$ can read the same table at the atomic types of $\mathcal{W}_A$. As in §3.3, an operation op notices the fold of A at operand values $\mathbf{U}$ if $\mathrm{op}(\mathbf{U}^{(A)}) \neq \mathrm{op}(\mathbf{U})^{(A)}$, and Boolean operations never notice a fold.

View every FO^2 formula as defining a binary relation, as in §5; its quantifiers then act as the projections $\exists y\, \psi \equiv \psi; \mathbf{1}$ and $\exists x\, \psi \equiv \mathbf{1}; \psi$. By induction on the formula, every relation that an FO^2 formula defines on $\mathcal{W}_0$ has a type table: if X has one, then $(X;\mathbf{1})(u,v)$ holds exactly when S_X contains the atomic type of some pair with first entry u, and these types are determined by $\pi(u)$; symmetrically for $\mathbf{1}; X$.

Lemma 5.3 (Projections notice at most one fold)**.** *If X has a type table, then $X;\mathbf{1}$ and $\mathbf{1};X$ each notice the fold of at most one $A \subseteq [k]$.*

Proof. At a pair (u,v), the prediction of $X;\mathbf{1}$ holds exactly when S_X contains the atomic type in $\mathcal{W}_0$ of some pair with first entry u, and $X^{(A)};\mathbf{1}$ holds exactly when S_X contains the atomic type in $\mathcal{W}_A$ of some such pair. If $u \neq \hbar$, the fold changes at most one pair with first entry u, namely $(u,\hbar)$; since every type occurs at least twice among these pairs in $\mathcal{W}_0$, the two sets of types coincide. If $u = \hbar$, the fold removes exactly the types $(1, r', \varnothing, A)$ and adds none. Hence $X;\mathbf{1}$ notices the fold of A only if the types in S_X of pairs with first entry $\hbar$ are all of the form $(1, r', \varnothing, A)$, and there is at least one; this determines A. The argument for $\mathbf{1};X$ uses second entries. □

Proposition 1.7 (CoR versus FO^2)**.** *For every $k \geq 1$, the* CoR *equation $\mathbf{id} \cup T_k = \mathbf{1}$ is FO^2-definable, but every equivalent FO^2 sentence over finite structures has at least 2^k quantifiers. Every FO^2 formula defining T_k also has at least 2^k quantifiers, and this bound is attained.*

Proof. For the upper bounds, write $\vartheta_A(u) \coloneqq \bigwedge_{i\in A} P_i(u,u) \wedge \bigwedge_{i\in A^c} \neg P_i(u,u)$ for a variable u. The FO^2 formula

$$\bigvee_{A\subseteq[k]} [\vartheta_A(y) \wedge \exists y\,(R(x,y) \wedge \vartheta_A(y))]$$

defines T_k with exactly 2^k quantifiers; the inner y is rebound only in the second conjunct. The equation $\mathbf{id} \cup T_k = \mathbf{1}$ says that for all x and all $y \neq x$, some R-successor of x has the loop pattern of y, and so does the FO^2 sentence

$$\forall x \bigwedge_{A\subseteq[k]} [\exists y\,(\neg x = y \wedge \vartheta_A(y)) \to \exists y\,(R(x,y) \wedge \vartheta_A(y))]\,.$$

For the lower bounds, every vertex of $\mathcal{W}_0$ has an R-successor with each loop pattern, so T_k is the full relation on $\mathcal{W}_0$ and the equation holds there. On $\mathcal{W}_A$, the hub has no R-successor with loop pattern A, so $T_k(\hbar, v)$ fails for every $v \neq \hbar$ with $\pi(v) = A$, and the equation fails. Let ψ be an FO^2 formula defining T_k, or an FO^2 sentence equivalent to the equation. Its value on $\mathcal{W}_0$ is the full relation, whose prediction on $\mathcal{W}_A$ is again full, but its value on $\mathcal{W}_A$ is not. If no quantifier of ψ noticed the fold of A, then, as in the proof of Lemma 3.3, every subformula would take its predicted value on $\mathcal{W}_A$, by induction on the formula: atomic types determine atomic formulas, and Boolean operations notice nothing. Hence some quantifier notices the fold of each A, and by Lemma 5.3 each quantifier notices at most one, so ψ has at least 2^k quantifiers. □

The structures have $11 \cdot 2^k$ vertices. Every CoR term translates linearly into an FO^3 formula, by translating composition as existential quantification and reusing the third variable at each composition. Hence the sentence $\forall x\,\forall y\,(x = y \vee \varphi_k(x,y))$, of size $O(k)$, also separates FO^3 from FO^2. Its existential counterpart, in contrast, collapses: $\mathbf{1}; T_k; \mathbf{1} = \mathbf{1}; R; \mathbf{1}$, since every edge $R(x,z)$ gives $T_k(x,z)$.

6 Scalar aggregation and matrix query languages

Relational composition aggregates two edge values over a common index. The calculation of §3.4 extends to scalar aggregation without any order or monotonicity assumption on the operation combining those values. This gives arity-elimination lower bounds for matrix query languages.

6.1 Finite join breadth and arbitrary kernels

Let $(S, \vee, 0_S)$ be a join-semilattice with least element 0_S. Its join breadth β is the least nonnegative integer such that every finite join equals the join of at most β of its summands. The empty join is 0_S. We assume $\beta < \infty$. A totally ordered set with maximum as join has breadth one. The space $\{0,1\}^b$ with coordinatewise disjunction has breadth b.

Let V be the common finite index set of the input matrices, with $n \coloneqq |V|$. For any function $\kappa : S \times S \to S$, called a kernel, define

$$(X ;_\kappa Y)(u,v) \coloneqq \bigvee_{z\in V} \kappa(X(u,z), Y(z,v)).$$

Kernels may differ between gates. They need not be associative, monotone, or zero-preserving. We also allow arbitrary uniform entrywise functions, transpose, uniform scalar constants, and a diagonal

mask with two fixed values, one on and one off the diagonal. "Uniform" means the same function is used at all positions of a given type; it has no access to vertex names. The inputs to the query are square $n \times n$ arrays. No order, vertex identifiers, arbitrary input vectors, or vertex-dependent constant arrays are supplied in addition to them.

Each circuit has a fixed directed acyclic wiring. Every entrywise function, kernel, and scalar constant is chosen before seeing the input; it may depend on n and k in a size-specific circuit. An entrywise function receives the scalar operands at its position, and a kernel receives only the two scalar arguments displayed above. In particular, a kernel does not also receive $X(u, v)$, the position names u, v, z, or an encoding of the whole input unless such a value has been computed and supplied through its two operands by permitted operations. Data-dependent values and tests are allowed, but there is no primitive for data-dependent rewiring or indexed access to arbitrary entries. These conventions specify the target model used in Theorem 1.8.

Reference arrays have scalar color tables, with entries $X[\mathbf{id}]$ and $X[\mu]$ in S, just as in §3.2, and predictions read these tables on the recolored graph. Entrywise functions and transpose preserve color tables and notice no fold. For arrays X, Y with color tables and $J \subseteq \Lambda$, define

$$\kappa_J^{\mathrm{L}} \coloneqq \bigvee_{\mu \in J,\, \nu \in \Lambda} \kappa(X[\mu], Y[\nu]), \qquad \kappa_J^{\mathrm{R}} \coloneqq \bigvee_{\mu \in J,\, \nu \in \Lambda} \kappa(X[\nu], Y[\mu]),$$
$$\kappa_J^{\mathrm{D}} \coloneqq \bigvee_{\mu \in J} \kappa(X[\mu], Y[\mu]), \qquad \kappa_* \coloneqq \kappa_\Lambda^{\mathrm{L}} = \kappa_\Lambda^{\mathrm{R}}.$$

The reference product $Z \coloneqq X \mathbin{;_\kappa} Y$ has the table

$$Z[\mathbf{id}] = \kappa(X[\mathbf{id}], Y[\mathbf{id}]) \vee \kappa_\Lambda^{\mathrm{D}},$$
$$Z[\mu] = \kappa(X[\mathbf{id}], Y[\mu]) \vee \kappa(X[\mu], Y[\mathbf{id}]) \vee \kappa_*.$$

The calculation of §3.4 remains valid, with endpoint contributions formed by κ, outside-witness contribution κ_* away from the hub, and the three possibly altered contributions $\kappa_J^{\mathrm{L}}, \kappa_J^{\mathrm{R}}, \kappa_J^{\mathrm{D}}$ at the hub row, column, and diagonal, respectively. This follows from the same color pairs of outside witnesses; idempotence of join makes multiplicities irrelevant.

Lemma 6.1 (Scalar local bounds)**.** *A scalar aggregation gate notices at most 3β folds. If one operand is constant across off-diagonal colors, it notices at most β; if both are, it notices none. For Boolean scalars with arbitrary kernels, the general bound improves to two. For b-bit scalars with coordinatewise disjunction and arbitrary kernels it improves to $2b$.*

Proof. We argue as for Lemma 3.4(b), with joins in place of the nonemptiness conditions. Choose at most β supporting summands for each of $\kappa_\Lambda^{\mathrm{L}}$, $\kappa_\Lambda^{\mathrm{R}}$, and $\kappa_\Lambda^{\mathrm{D}}$. From these summands retain, respectively, the row colors, the column colors, and the common colors on the diagonal. For example, a supporting summand of the first join chooses a pair (μ, ν), and only its first color μ must remain in J. Retaining their union preserves the product everywhere. Its size is at most 3β, and the gate can notice the fold of λ only if λ lies in this union.

If $Y[\mu]$ does not depend on μ, then $\kappa_J^{\mathrm{L}} = \kappa_J^{\mathrm{D}}$ and $\kappa_J^{\mathrm{R}} = \kappa_*$ for every nonempty J. Only one join needs support. If both tables are constant, all three joins are unchanged. For Boolean scalars, if $\kappa_\Lambda^{\mathrm{D}} = 1$, a single color attaining that value simultaneously supports both other joins. Otherwise the diagonal join is always zero, and at most one supporting color for each of the other joins suffices. Apply this argument independently to each output bit for the b-bit bound. □

These calculations show that color tables stay closed and bound the number of folds that each gate notices, as Theorem 3.5 requires; §6.3 supplies the query tests. Uniform entrywise functions, transpose, and the permitted constants notice no fold.

Remark 6.2 (Why three colors can be needed). The scalar coefficient three is attained locally even at breadth one. Use max–min on $[0, 2]$, and choose three distinct colors μ_1, μ_2, μ_3 with

$$(X[\mu_1], Y[\mu_1]) = (2, 0), \qquad (X[\mu_2], Y[\mu_2]) = (0, 2), \qquad (X[\mu_3], Y[\mu_3]) = (1, 1).$$

Set the other color values and both diagonal values to zero. The fold of μ_1 changes the hub-row value from 2 to 1; that of μ_2 changes the hub-column value from 2 to 1; that of μ_3 changes the hub-diagonal value from 1 to 0. Thus one product notices three folds. The three affected positions explain the scalar bound. Proposition C.2 gives a matching asymptotic product count for the full symmetric recoloring promise, and Appendix C.6 also gives the corresponding b-bit packing.

6.2 Typed matrix operations

We use the dimension symbols n and 1. Intermediate arrays have shapes $n \times n$, $n \times 1$, $1 \times n$, or 1×1; these are matrices, column vectors, row vectors, and scalars, respectively. An admissible contraction of an $a \times n$ and an $n \times c$ array is

$$Z(i, j) = \bigvee_{z=1}^{n} \kappa(X(i, z), Y(z, j)), \qquad a, c \in \{1, n\}.$$

Products contracting an index of size one are just applications of the kernel and are not data-dimension contractions. We allow entrywise functions, transpose, scalar and vector broadcasting, diagonal extraction and formation, uniform constant vectors, and the diagonal mask. All of these are uniform and do not aggregate a data index. This is at least as expressive as the corresponding MATLANG fragment with semiring multiplication as kernel; standard MATLANG operations are described in [BGVW19, BGV22].

On the reference structure, every vector is constant over its data index and every scalar is a single value. Encode a column vector $\mathbf{c}$ as the square array $(u, v) \mapsto \mathbf{c}(u)$, a row vector $\mathbf{r}$ as $(u, v) \mapsto \mathbf{r}(v)$, and a scalar as a constant square array. Reference closure follows from the color-table calculation. The encoded reference vector has the same value on every color and the diagonal, so it is its own prediction. Diagonal formation and extraction and all broadcasting operations notice no fold. Matrix–vector and vector–matrix contractions notice at most β folds by Lemma 6.1. A vector inner product notices none, since both operand values on the reference structure are constant and are their own predictions. Products over a size-one index notice none either.

Let $\#_{\mathrm{mv}}(C)$ count matrix–vector and vector–matrix contractions, and let $\#_{\mathrm{mm}}(C)$ count square matrix–matrix contractions, always counting shared gates once. The local bounds give the following typed resource estimate.

Lemma 6.3 (Cost of typed products)**.** *Every typed matrix circuit C in the model above satisfies*

$$\mathrm{cost}(C) \le \beta(\#_{\mathrm{mv}}(C) + 3\#_{\mathrm{mm}}(C)).$$

For Boolean scalars and arbitrary kernels, the bound improves to $\#_{\mathrm{mv}}(C) + 2\#_{\mathrm{mm}}(C)$; for b-bit scalars it is $b(\#_{\mathrm{mv}}(C) + 2\#_{\mathrm{mm}}(C))$.

Proof. Sum the bounds of Lemma 6.1 over the distinct product gates. The other permitted operations notice no fold, as established above. □

The gate-by-gate accounting can be summarized as follows.

Operation	Folds noticed, at most
Uniform entrywise function, transpose, permitted constant	0
Diagonal formation or extraction, broadcasting	0
Product over an index of size one	0
Vector inner product	0
Matrix–vector or vector–matrix contraction	β
Square matrix–matrix contraction	3β

The cost counts noticed folds; an operation that notices none may still take time to evaluate. Inner products contribute zero to the cost.

6.3 Arity elimination for a weighted query

Let $(S, \vee, \otimes, 0_S, 1_S)$ be a nontrivial commutative idempotent semiring of finite join breadth β. Thus addition $\vee$ is an idempotent commutative monoid, multiplication $\otimes$ is a commutative monoid, multiplication distributes over addition, and 0_S annihilates multiplication. For the common input index set V, define

$$\mathcal{G}_k(x, y) \coloneqq \bigvee_{z \in V} \bigotimes_{i=1}^{k} \left[(P_i(x, y) \otimes Q_i(x, z)) \vee (Q_i(x, y) \otimes P_i(x, z)) \right].$$

An annotated relation assigns a scalar in S to each tuple. In positive annotated relational algebra, union is pointwise addition, natural join multiplies annotations, and projection sums over removed attributes. Renaming changes attribute names. We write ARA(3) for this language restricted to intermediate relations with at most three attributes. The weighted query $\mathcal{G}_k$ has an ARA(3) expression of size $O(k)$: form the two ternary joins in each clause, take their union, join the clauses, and project out z.

Theorem 1.8 (Exponential arity-elimination overhead)**.** *For every such semiring and $k \geq 1$, any typed matrix circuit computing $\mathcal{G}_k$ on all finite inputs requires at least $2^k/(3\beta)$ matrix–vector, vector–matrix, or matrix–matrix aggregation gates in total. This holds with arbitrary entrywise functions, each applied in the same way at every entry, and arbitrary kernels in the model of §6.*

More precisely, writing $\#_{\mathrm{mv}}(C)$ and $\#_{\mathrm{mm}}(C)$ for the typed product counts just defined, every exact circuit satisfies

$$\beta(\#_{\mathrm{mv}}(C) + 3\#_{\mathrm{mm}}(C)) \geq 2^k.$$

For Boolean scalars the stronger inequality is $\#_{\mathrm{mv}}(C) + 2\#_{\mathrm{mm}}(C) \geq 2^k$. Each inequality holds separately at every $n \geq n_0(2^k)$. For an input-independent randomized circuit with pointwise error at most $\varepsilon < 1/2$, the corresponding assertions are

$$\beta\,\mathbb{E}(\#_{\mathrm{mv}}(C) + 3\#_{\mathrm{mm}}(C)) \geq (1 - 2\varepsilon)2^k,$$
$$\mathbb{E}(\#_{\mathrm{mv}}(C) + 2\#_{\mathrm{mm}}(C)) \geq (1 - 2\varepsilon)2^k \quad \text{(Boolean scalars)}.$$

For every fixed semiring in the theorem, the optimal total circuit size is $\Theta(2^k)$.

Proof. Use the $0_S, 1_S$ encodings of the Boolean G_k tests. Nontriviality makes these values distinct, and idempotence makes the value of $\mathcal{G}_k$ exactly its Boolean truth value under this encoding. Thus there are 2^k tests. Apply Theorem 3.5 and Lemma 6.3, or its Boolean refinement. Lemma 3.1 provides the construction at every stated size, and the expected-cost part of the theorem gives the randomized inequalities.

The upper bound uses exactly 2^k matrix–vector contractions. Define H_A by replacing the intersections in its earlier definition by pointwise products $\otimes$. For each A, contract H_{A^c} with the constant 1_S column vector, broadcast the result along rows, multiply entrywise by H_A, and join the outputs over A. Distributivity and commutativity prove that this computes $\mathcal{G}_k$ on arbitrary inputs. All H_A and H_{A^c} can be shared with $O(2^k)$ total nodes. Hence for a fixed semiring of finite breadth the minimum circuit size is $\Theta(2^k)$, and the positive ARA(3) source has linear size. □

In the Boolean one-sided model, mixing this exact computation with the constant-zero and constant-one outputs attains the randomized coefficient $1 - 2\varepsilon$, as in §4.3.

Examples of breadth-one semirings include max–min on $[0, 1]$, $(\mathbb{R} \cup \{-\infty\}, \max, +, -\infty, 0)$, and $(\mathbb{R} \cup \{+\infty\}, \min, +, +\infty, 0)$. For each, at least $2^k/3$ data-dimension products are necessary, even in the enlarged kernel model. Propositions D.1 and D.2 in Appendix D give complementary bounds for the disjunctive query F_k over finite chains and tropical scalars.

6.4 Positive computations without idempotence

A separate transfer handles positive arithmetic, for which join breadth is not available. Let $(S, +, \otimes, 0_S, 1_S)$ be a nontrivial commutative semiring satisfying

$$r\,1_S \neq 0_S \qquad \text{for every integer } r \geq 1, \tag{NV}$$

where $r\,1_S$ is the sum of r copies of 1_S. Restrict the target to positive entrywise addition and multiplication, semiring matrix products, transpose, broadcasting, diagonal operations, and constants in $\mathbb{N}1_S := \{r\,1_S : r \geq 0\}$. Arbitrary entrywise decoding and negative constants are not allowed here. Replace joins in $\mathcal{G}_k$ by sums to define its positive semiring version.

On Boolean-encoded inputs, every intermediate belongs to $\mathbb{N}1_S$. The map sending 0_S to zero and every other element of this subsemiring to one is a homomorphism to the Boolean semiring. Indeed, sums and products of positive multiples of 1_S are again nonzero by the nonvanishing condition (NV). Applying this map to every gate of a target circuit C gives a Boolean matrix circuit with exactly the same typed product counts. Its output is G_k. The Boolean lower bound therefore gives $\#_{\mathrm{mv}}(C) + 2\#_{\mathrm{mm}}(C) \geq 2^k$. For a randomized circuit, correctness of a scalar output implies correctness after this map, so the same transfer gives $\mathbb{E}(\#_{\mathrm{mv}}(C) + 2\#_{\mathrm{mm}}(C)) \geq (1 - 2\varepsilon)2^k$. Here the cost is applied to the Booleanized computation; no argument by predictions is claimed for the original scalar circuit. The 2^k-product expansion still works by distributivity. This includes positive computations over $\mathbb{N}$ and nonnegative reals.

Appendix C.6 shows why the operation restrictions matter: arbitrary decoding together with nonidempotent summation can encode the counts of all profiles in a row in one aggregation. Conversely, the arbitrary-kernel result of Theorem 1.8 is valid because finite join breadth limits how many colors can support a single aggregation.

7 Open questions

Our bounds settle exponential FO^3-to-CoR compilation overhead over growing binary signatures. Several quantitative questions remain.

A fixed signature. Does a superpolynomial gap persist for full CoR over a fixed finite signature, or even one binary relation symbol? The encoding of [Nak22, §6] gives a superpolynomial separation for the existential fragment over a single symbol. Conditional evidence points the same way for full CoR: under the 3-OV hypothesis, the argument of [NU27, Corollary 40] excludes polynomial-time translations already for existential positive FO^3 sentences over one binary relation symbol [NU27, Theorem 39]. Extending the unconditional conclusion to full CoR requires controlling unrestricted complement on the encoded inputs. Our present formulas use one or two binary symbols per coordinate.

Domain size. Proposition 1.3 places the onset of the blowup between domain sizes $2^k/(8k)$ and $O(k4^k)$, and leaves $\Omega(\sqrt{n/\log n}) \le c_n(G_k) \le 4kn$ when $n < n_0(2^k)$. The lower bound is limited by the $2N^2 + 2$ vertices that two witnesses for N colors require, and the upper bound uses only row projections. Closing the gap between $\sqrt{n/\log n}$ and kn may require a different reference structure or compositions that combine different rows.

The exact composition count for G_k. On arbitrary finite inputs the optimum lies between 2^{k-1} and 2^k; without converse it is exactly 2^k. Proposition C.1 attains 2^{k-1} on every input in the symmetric hub-recoloring family. Consequently that family alone cannot improve the full-CoR lower bound. Determining the optimum on arbitrary inputs calls for either a more efficient general circuit or a family exposing an additional obstruction.

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A Explicit two-witness colorings

This appendix gives two deterministic versions of Lemma 3.1. The affine-plane construction follows the direction-partition method of Jipsen, Maddux, and Tuza [JMT95]. Their assignment of at least two directions to every color already guarantees two witnesses at plane order at least four; we include the verification because this is the property used here. The second construction retains the all-sufficiently-large-sizes guarantee of the random proof.

Proposition A.1 (Explicit two-witness colorings)**.** *For every palette size $N \geq 2$, a graph with the two-witness property of Lemma 3.1 can be constructed deterministically on fewer than $16N^2$ vertices. For every prescribed $n \geq n_0(N)$, such a graph on exactly n vertices can be constructed in time polynomial in n, N.*

We prove the two assertions in the following subsections.

A.1 A quadratic-size affine-plane construction

Let s be the smallest power of two with $s \geq 2N - 1$, and let $\mathbb{F}_s$ be the field of s elements. Use vertex set $V := \mathbb{F}_s^2$. There are $s + 1$ directions of affine lines. Partition these directions into sets Dir_μ, $\mu \in \Lambda$, each of size at least two. Color distinct points u, v by the unique μ containing the direction of their joining line. Since $N \geq 2$, we have $s \geq 4$ and $|V| = s^2 < 16N^2$.

Fix distinct u, v and requested colors μ, ν. If $\mu = \nu = \chi(u, v)$, all $s - 2 \geq 2$ other points of the line uv are witnesses. Otherwise, choose a direction from Dir_μ at u and a direction from Dir_ν at v, neither equal to the direction of uv, and choose the two directions to be distinct. Their two lines intersect at a unique witness outside $\{u, v\}$. There are at least two choices: if the colors differ, their direction sets are disjoint and at most one set loses a direction to uv; if they agree but differ from $\chi(u, v)$, two directions from their common set can be taken in either order. Distinct direction pairs give distinct intersection points, since a point outside $\{u, v\}$ determines both directions uniquely. Thus the coloring has the two-witness property. For $N = 2^k$, one can choose $s = 2N$, giving 4^{k+1} vertices. This construction supplies selected domain sizes, rather than every large size; induced subgraphs need not retain the extension property.

A.2 Every sufficiently large size

Fix any $n \geq n_0(N)$ and start with independently uniform edge colors. For distinct u, v, w and colors μ, ν, let $\mathcal{B}_{u,v,w,\mu,\nu}$ be the event that there is no witness outside $\{u, v, w\}$. Their expected total number is less than one by the proof of Lemma 3.1. Under any partial assignment σ of edge colors, independence of the remaining edges gives

$$\Pr(\mathcal{B}_{u,v,w,\mu,\nu} \mid \sigma) = \prod_{z \notin \{u,v,w\}} (1 - \Pr(\chi(u,z) = \mu \mid \sigma) \Pr(\chi(v,z) = \nu \mid \sigma)) .$$

Each factor uses probabilities in $\{0, 1/N, 1\}$. At the next uncolored edge, evaluate the sum of these conditional probabilities for each of its N possible colors and choose a color minimizing the sum. The average of the choices is the current sum, so this never increases it. After all edges are assigned, the number of bad events is an integer strictly below one, hence zero. This is a deterministic algorithm polynomial in n, N: there are polynomially many events, factors, edge choices, and bits in their rational representations. The same argument applies to the directed construction in Appendix E.1, with independently colored ordered arcs.

B A fixed circuit for the Hansel bound

We prove Proposition 4.4. Hansel's monotone identification bound [Han66, AS17] concerns adaptive membership queries. Here the queried set may differ between rows, so we spell out the fixed wiring and its total number of Boolean gates. All notation in this appendix is local: Γ is a family of chains, $\gamma = (W_0 \subset \cdots \subset W_{|\gamma|-1})$ is one chain, and the auxiliary subsets V_i below are not vertex sets.

B.1 The symmetric chains

Identify a subset of $[k]$ with its bit string. Treat zero as an opening parenthesis and one as a closing parenthesis, and match them by the usual stack procedure. Unmatched positions have the pattern $1 \cdots 1 0 \cdots 0$. Hold all matched pairs fixed and let the unmatched positions run through all patterns of this form. This partitions the Boolean cube into saturated chains, each symmetric about rank $k/2$. Denote their family by Γ. Every chain meets rank $\lfloor k/2 \rfloor$ exactly once, so $|\Gamma| = B_k$. The number of chains of length at least two is $\binom{k}{\lfloor k/2 \rfloor + 1}$: for odd k every chain has even length; for even k, these are exactly the chains meeting the rank above the middle.

For $0 \leq i \leq |\gamma| - 3$ define

$$V_i \coloneqq W_i \cup (W_{i+2} \setminus W_{i+1}).$$

Then $W_i \subset V_i \subset W_{i+2}$. The two newly affected unmatched positions form a new matched 01 pair; all previous pairs remain matched. Consequently V_i lies in a chain of length $|\gamma| - 2$. This is the chain property needed below. Process the chains in increasing order of length, so answers for every V_i have already been computed.

B.2 Two row queries per chain

For each subset $W \subseteq [k]$, let

$$E_W \coloneqq R_W; \mathbf{1}.$$

Its value in a row says whether some entry in that row satisfies all relations with indices in W. These are row-constant relations, and $W \subseteq W'$ implies $E_{W'} \subseteq E_W$. For a chain γ, put $a_i \coloneqq E_{V_i}$, and use Boolean operations on row-constant relations to define

$$l_j \coloneqq \bigcup_{i \geq j} a_i, \qquad u_j \coloneqq \bigcap_{i \leq j-2} a_i, \qquad U_j \coloneqq u_j \cap \overline{l_j} \quad (0 \leq j < |\gamma|).$$

Indices i in these operations range from 0 to $|\gamma| - 3$; empty unions and intersections have their usual meanings. The chain property implies $l_j \subseteq E_{W_j} \subseteq u_j$. In any fixed row, the positions with $U_j = 1$ form an interval containing at most two adjacent positions. Indeed l_j and u_j are nonincreasing in j, and having both j and $j+2$ unknown would require $a_j = 0$ from $l_j = 0$ and $a_j = 1$ from $u_{j+2} = 1$.

Put $U_{-1} \coloneqq \mathbf{0}$ and $U_{|\gamma|} \coloneqq \mathbf{0}$, and define the two selectors

$$\mathrm{fst}_j \coloneqq U_j \cap \overline{U_{j-1}}, \qquad \mathrm{lst}_j \coloneqq U_j \cap \overline{U_{j+1}}.$$

At most one first and one last position is selected in each row. Compute

$$Z_{\mathrm{fst}} \coloneqq \left(\bigcup_j (\mathrm{fst}_j \cap R_{W_j}) \right) ; \mathbf{1}, \qquad Z_{\mathrm{lst}} \coloneqq \left(\bigcup_j (\mathrm{lst}_j \cap R_{W_j}) \right) ; \mathbf{1}.$$

Because the selectors depend only on the row, these products ask exactly the two selected membership queries, with no cross-talk between rows. Recover all answers by

$$E_{W_j} = l_j \cup (\mathrm{fst}_j \cap Z_{\mathrm{fst}}) \cup (\mathrm{lst}_j \cap Z_{\mathrm{lst}}).$$

For a singleton chain only the first query is needed. All other operations here are fixed Boolean gates; the circuit has no adaptive wiring.

Finally output

$$\bigcup_{W \subseteq [k]} (R_{W^c} \cap E_W),$$

where $W^c \coloneqq [k] \setminus W$. This is the expansion of F_k. The number of row projections is at most

$$B_k + \binom{k}{\lfloor k/2 \rfloor + 1} = B_{k+1}.$$

Precompute all R_W by sharing intersections along a subset tree. Within a chain, suffix unions and prefix intersections compute all l_j, u_j with $O(|\gamma|)$ nodes, and the selectors, queries, and reconstruction take another $O(|\gamma|)$ nodes. Since $\sum_{\gamma \in \Gamma} |\gamma| = 2^k$, the total circuit size is $O(2^k)$. A query for a chain depends only on chains at least two shorter, so any path crosses at most $\lfloor k/2 \rfloor + 1$ projection gates.

Replacing row projections by rebinding y also gives a FO^2 formula circuit with at most B_{k+1} distinct quantifier gates and $O(2^k)$ total nodes. Expanding the shared definitions may duplicate queries; this is not an upper bound of B_{k+1} on quantifier occurrences of an ordinary formula, or on composition occurrences of a CoR term.

C Tree bounds and limits of the method

C.1 The input-occurrence charging lemma

This subsection proves the tree refinement stated in §5.2.

Lemma 5.2 (Disjoint input-occurrence charging)**.** *Fix the reference structure and the test folds of §3.5. Suppose every singleton selector of a test color requires at least $d > 0$ input-relation occurrences. Then every* CoR *term T, whether correct or not, satisfies*

$$\mathrm{cost}(T) \le \#_{\mathrm{rel}}(T)/d.$$

Proof. For a subterm whose value on the reference structure is X, let $S_X \subseteq \Lambda$ be the off-diagonal support of X. Boolean operations act on these sets in the usual way and converse fixes them. By the color-table identities (CT), a composition $X;Y$ with both supports nonempty has off-diagonal support Λ. Call such a node **active**. If either support is empty, the off-diagonal table of the product is zero or one of the two operand tables, according to the diagonal bits. Such an inactive node notices no fold: every outside-witness contribution in the local calculation vanishes.

Construct a Boolean formula for the off-diagonal table of any subterm by replacing active compositions by the constant true, inactive compositions by zero or the relevant operand table, and converse by its argument. This operation never duplicates an input occurrence. For each active node $g = (X;Y)$, apply this pruning to its two operands. Let $\mathcal{L}_g$ be the set of original input occurrences that remain in the two resulting table formulas. These are the occurrences that *survive the pruning*, not all input occurrences below g. This distinction prevents repeated charging at successive active nodes. The sets $\mathcal{L}_g$ are pairwise disjoint: an active descendant is replaced by a constant when an ancestor's operands are pruned, and incomparable nodes have disjoint subtrees.

By Lemma 3.4(a), a composition notices the fold of a test color λ only if

$$S_X = \{\lambda\}, \qquad S_Y = \{\lambda\}, \qquad \text{or} \qquad S_X \cap S_Y = \{\lambda\}.$$

Let E_g be the set of test colors satisfying at least one of these conditions. Thus $|D_g| \le |E_g|$. If $S_X \cap S_Y$ is nonempty, E_g has size at most one. Its unique test color, if present, is selected by one operand table or by their conjunction. The selector assumption therefore gives $|\mathcal{L}_g| \ge d|E_g|$. If the two supports are disjoint, only singleton operands contribute to E_g. Each contributing operand has at least d input occurrences, and the operands are disjoint trees. The same inequality follows. Summing over active nodes proves

$$\mathrm{cost}(T) = \sum_g |D_g| \le \sum_g |E_g| \le \frac{1}{d} \sum_g |\mathcal{L}_g| \le \#_{\mathrm{rel}}(T)/d.$$

Diagonal bits may affect which operand table an inactive node retains, but do not create additional input occurrences in this argument. □

C.2 Factored trees for G_k

A complete binary decision tree successively chooses P_i or Q_i for the read edge. At leaf A place $H_{A^c}; \mathbf{1}$. A branch at coordinate i with child terms U_1, U_0 has the form

$$(P_i \cap U_1) \cup (Q_i \cap U_0).$$

Distributing the branches recovers the DNF for G_k on arbitrary inputs; no assumption that exactly one of P_i, Q_i holds at each edge is used. Each of the 2^k leaves costs $2k+1$ nodes, and each of the 2^k-1 internal branches adds five. The total is $(2k+6)2^k - 5$. The number of relation-input occurrences is $k2^k + 2(2^k - 1)$. Replacing each leaf by $\exists y\, H_{A^c}(x, y)$ gives a FO^2 tree with $(2k+5)2^k - 5$ nodes.

C.3 Why the requirements are crossed

For comparison, the superficially similar non-crossed formula collapses:

$$\exists z \bigwedge_{i=1}^{k} [(P_i(x,y) \wedge P_i(x,z)) \vee (Q_i(x,y) \wedge Q_i(x,z))] \equiv \bigwedge_{i=1}^{k} (P_i(x,y) \vee Q_i(x,y)).$$

The forward implication is immediate; for the reverse implication take $z = y$. In G_k, the two edges must meet different requirements at every coordinate, so the choice $z = y$ no longer works.

C.4 The antichain limit for Boolean F_k

Let R_i hold on color μ exactly when $i \in \mu$, as in §4.2, for an arbitrary palette of subsets of $[k]$. A read edge of color λ^c then accepts every witness color containing λ. To distinguish the removal of a single accepted color, that color must be the unique accepted palette color. In particular, all such test colors are inclusion-maximal palette colors and form an antichain. An antichain $\mathcal{H} \subseteq 2^{[k]}$ has size at most B_k: a uniformly random ordering of $[k]$ has a given λ as an initial set with probability $1/\binom{k}{|\lambda|}$, and at most one member of an antichain can occur among its initial sets. Thus $\sum_{\lambda \in \mathcal{H}} \binom{k}{|\lambda|}^{-1} \leq 1$ and $|\mathcal{H}| \leq B_k$. This is a limit of fold tests with such interpretations, not an upper bound on every possible lower-bound method for F_k.

C.5 Converse on the promised symmetric family

Proposition C.1 (Sharpness on the symmetric recoloring family)**.** *For every $k \geq 1$, on the reference structure over the cube and all its recolorings at the hub by maps $f : \Lambda \to \Lambda$, the minimum number of composition gates in a full-*CoR *circuit computing G_k is 2^{k-1}.*

Proof. Let $N := 2^k$ and retain the symmetric reference structure over the cube. For each unordered complementary pair $\{A, A^c\}$ choose one representative A and compute, once,

$$X_A := H_A; H_{A^c}.$$

Use the shared result in

$$\bigcup_{\{A,A^c\}} \left[(H_{A^c} \cap X_A) \cup (H_A \cap X_A^{\smile})\right].$$

This circuit has exactly $N/2$ compositions and computes G_k on every input obtained by recoloring all edges at the fixed hub by an arbitrary map f. To check this, write $J := f(\Lambda)$. The relation X_A is true on all distinct non-hub pairs, has zero diagonal, and has hub-row and hub-column values respectively $[A \in J]$ and $[A^c \in J]$. The masks select the complementary read colors. At a non-hub vertex every color still occurs, so G_k is true on every off-diagonal read edge there. At the hub it is true precisely when the complementary color lies in J. These are exactly the displayed circuit's values. The lower bound follows by applying Theorem 3.5 and Lemma 3.6 to the N folds in this family. Thus the circuit attains the optimum on the promised family. □

C.6 Scalar packing and the role of the operation model

The Boolean promised-family example shows that a composition can notice two folds. The next example explains the three scalar observations at the hub. It concerns the same promise, not an upper bound on arbitrary input structures.

Proposition C.2 (Three scalar observations)**.** *Let $N \coloneqq 2^k$, with $k \geq 1$. On the symmetric reference structure over the cube and all its recolorings at the hub by arbitrary maps $f : \Lambda \to \Lambda$, the Boolean query G_k can be computed with at most $\lceil N/3 \rceil + 1$ max–min products over $[0, 2]$, using arbitrary uniform entrywise functions, transpose, and diagonal operations.*

Proof. Put $m \coloneqq \lceil N/3 \rceil$. Choose m distinct complementary pairs $\{A_i, A_i^c\}$, and assign each of the remaining $r \coloneqq N - 2m \leq m$ colors to a different pair. These choices are possible for $N = 2^k \geq 2$. Call an assigned residual color ρ_i. The singleton selectors H_A and entrywise functions construct arrays X_i, Y_i whose off-diagonal values $(X_i[\mu], Y_i[\mu])$ are

$$(2, 0) \text{ at } A_i, \quad (0, 2) \text{ at } A_i^c, \quad (1, 1) \text{ at } \rho_i \text{ if assigned}, \quad (0, 0) \text{ otherwise.}$$

Put both diagonal values zero and compute one max–min product S_i. Write $J \coloneqq f(\Lambda)$ for the colors still present at the hub. The value of S_i at distinct non-hub vertices is two. At the hub, the local product calculation gives

$$[S_i(\hbar, v) = 2] = [A_i \in J], \qquad [S_i(u, \hbar) = 2] = [A_i^c \in J], \qquad S_i(\hbar, \hbar) = [\rho_i \in J]$$

for non-hub u, v, with the last equality only for assigned pairs. The diagonal value at any non-hub vertex is one for such a pair. Read masks $H_{A_i^c}, H_{A_i}$, together with S_i and its transpose, therefore recover the answers for both selected colors on every non-loop read edge. All loops are suppressed by these masks.

It remains to recover the residual colors. For each assigned ρ_i, diagonal extraction followed by an entrywise test gives a vector

$$b_i(u) = [S_i(u, u) < 1].$$

It is zero outside $\hbar$, and at $\hbar$ it records that ρ_i is missing. Encode these finitely many bits in distinct binary positions of a single nonnegative scalar, rescaled into $[0, 2]$, at each diagonal entry of an array. One row projection, a max–min product with the constant-two array, broadcasts this code along its row. A fixed entrywise decoder recovers each bit. Intersecting its complement with the read mask $H_{\rho_i^c}$ supplies exactly the answers for witness color ρ_i. At non-hub vertices every color is present and all decoded missing bits are zero, as required. Combining these outputs with those for the selected pairs computes G_k on the entire promised family. The construction uses m products S_i and at most one final projection. □

For a b-bit state with coordinatewise disjunction as join, b tests can instead be packed into one row projection. On arbitrary Boolean inputs, group the 2^k DNF witness tests into blocks of at most b, aggregate their truth values in parallel, and decode and mask the resulting bits entrywise. This uses $\lceil 2^k/b \rceil$ projections. It complements the lower bound $b(\#_{\mathrm{mv}}(C) + 2\#_{\mathrm{mm}}(C)) \geq 2^k$ of §6.

Finite join breadth, or a positive-semiring restriction, cannot simply be removed. With ordinary real addition and arbitrary entrywise decoding, all finite row histograms of Boolean profiles (§4.4) can be encoded in one projection: for an n-element domain, encode profile number j by $(n+1)^j$ and

sum along the row. Base-$(n+1)$ digits recover every profile count. A read-edge-dependent decoder then evaluates G_k, and likewise F_k with profiles listing the values of $R_1, \dots, R_k$. This example allows arbitrary nonpositive entrywise functions and thus lies outside the positive fragment of §6.4.

D Weighted disjunctive queries

This appendix proves Propositions D.1 and D.2, supplementary applications of the common test–cost theorem to finite-chain and tropical versions of F_k. The notation in this appendix is local: ℓ is the number of positive levels in a chain, and $t_{k,\ell}, \widehat{t}_{k,\ell}$ are the lower- and upper-bound counts defined below. Throughout $k \geq 2$.

D.1 A finite-chain transition

On $S = \{0, 1, \dots, \ell\}$, with $\ell \geq 1$, use maximum as addition and minimum as multiplication, and define

$$\mathcal{F}_{k,\ell}(x, y) \coloneqq \max_{z} \min_{i \in [k]} \max(R_i(x, y), R_i(x, z)).$$

Put

$$t_{k,\ell} \coloneqq \sum_{j=\lceil k/2 \rceil}^{\min(k, \lceil k/2 \rceil + \ell - 1)} \binom{k}{j},$$
$$\widehat{t}_{k,\ell} \coloneqq \sum_{j=\max(0, \lfloor k/2 \rfloor - \ell + 1)}^{\min(k, \lfloor k/2 \rfloor + \ell)} \binom{k}{j}.$$

Symmetry and unimodality of binomial coefficients give $\widehat{t}_{k,\ell} \leq 2t_{k,\ell}$. They also give

$$t_{k,\ell} = \Theta\Big(\min\{2^k, \ell B_k\}\Big),$$

with absolute constants: the upper bound follows by counting levels and all subsets, and the first $\min\{\ell, \lfloor \sqrt{k} \rfloor\}$ levels above the middle each have size at least a constant multiple of B_k. For small k the constants can be adjusted uniformly.

Proposition D.1 (Finite-chain disjunctive queries)**.** *For $k \geq 2$ and $\ell \geq 1$, the minimum number of matrix–vector, vector–matrix, and square matrix–matrix products needed to compute $\mathcal{F}_{k,\ell}$ on all finite inputs is $\Theta(t_{k,\ell})$, with absolute constants. This holds in the enlarged max-aggregation model with arbitrary uniform entrywise functions and two-input kernels. More precisely, every correct circuit satisfies $\#_{\mathrm{mv}}(C) + 3\#_{\mathrm{mm}}(C) \geq t_{k,\ell}$. There is an exact circuit using at most $\widehat{t}_{k,\ell} \leq 2t_{k,\ell}$ row projections.*

Proof. Let $\mathcal{H}$ consist of the ℓ upper levels in the definition of $t_{k,\ell}$, truncated at k, and let $\Lambda \coloneqq \mathcal{H} \cup \{\mu^c : \mu \in \mathcal{H}\}$. For a color $\mu \in \Lambda$, assign the amplitude

$$a(\mu) \coloneqq \begin{cases} \ell - (|\mu| - \lceil k/2 \rceil), & \mu \in \mathcal{H}, \\ \ell, & \mu \notin \mathcal{H}. \end{cases}$$

Set $R_i(\mu) \coloneqq a(\mu)$ when $i \in \mu$ and zero otherwise, and put all loop values zero. For a test color $\lambda \in \mathcal{H}$, use read color λ^c and the fold of λ. The read values are ℓ on λ^c and zero on λ. A witness color contributes zero unless it contains λ. Color λ contributes $a(\lambda) > 0$, and every strict superset in the palette contributes a smaller amplitude. Loops contribute zero. Thus the value strictly decreases when color λ disappears from the hub row. There are $t_{k,\ell}$ tests. Theorem 3.5 gives

$$\#_{\mathrm{mv}}(C) + 3\#_{\mathrm{mm}}(C) \geq t_{k,\ell}$$

for the enlarged max-aggregation model with arbitrary kernels, and $(1 - 2\varepsilon)t_{k,\ell}$ on the right for expected randomized counts. For $\ell = 1$, the Boolean bound improves the coefficient three to two. For a fixed domain size it suffices that $n \geq n_0(|\Lambda|)$.

For completeness, a finite-valued version of the circuit in Appendix B uses at most $\widehat{t}_{k,\ell}$ row projections. Now put $R_W \coloneqq \min_{i \in W} R_i$, with $R_\varnothing \coloneqq \ell$, and $E_W(x) \coloneqq \max_z R_W(x, z)$. Use the same chains and auxiliary sets V_i. With $a_i \coloneqq E_{V_i}$, the bounds

$$l_j = \max_{i \geq j} a_i \leq E_{W_j} \leq u_j = \min_{i \leq j-2} a_i$$

hold, with empty maximum zero and empty minimum ℓ. At each threshold $\eta \in \{1, \ldots, \ell\}$, the Boolean proof shows at most two unknown positions. Hence at most 2ℓ positions satisfy $l_j < u_j$ in any row. Put $s_\gamma \coloneqq \min\{|\gamma|, 2\ell\}$. For $1 \leq r \leq s_\gamma$, let the row-constant selector $\sigma_{r,j}(x)$ be ℓ if j is the rth unknown position in row x, and zero otherwise. A fixed prefix computation implements these selectors. Compute one row projection for each r:

$$Z_r(x) \coloneqq \max_z \max_{0 \leq j < |\gamma|} \min\{\sigma_{r,j}(x), R_{W_j}(x, z)\}.$$

In each row at most one j is selected, so $Z_r(x)$ is exactly its membership-query answer, or zero when that row has no rth unknown position. Recover every answer by

$$E_{W_j}(x) = \max\left\{ l_j(x),\ \max_{1 \leq r \leq s_\gamma} \min\{\sigma_{r,j}(x), Z_r(x)\} \right\}.$$

For an unknown position the selected answer lies between l_j and u_j; for a known position all selectors vanish and $l_j = u_j = E_{W_j}$. A prefix computation of the number of unknown positions, using one-hot Boolean states up to $\min\{|\gamma|, 2\ell\}$, implements all these selectors with $O(\min\{\ell, k\}|\gamma|)$ gates. No data-dependent wiring is required.

A chain needs at most $\min\{|\gamma|, 2\ell\}$ queries. By symmetry this is its number of elements in the central 2ℓ ranks used to define $\widehat{t}_{k,\ell}$. Summing over chains gives the stated query count, $O(\min\{\ell, k\}2^k)$ total nodes, and composition depth at most $\lfloor k/2 \rfloor + 1$. The output is $\max_{W \subseteq [k]} \min(R_{W^c}, E_W)$, equal to $\mathcal{F}_{k,\ell}$ by distributivity. The direct expansion alternatively uses at most $2^k - 1$ projections and $O(2^k)$ shared nodes. Consequently the optimal number of data-dimension products is $\Theta(t_{k,\ell})$. When $\ell = \Omega(\sqrt{k})$, total circuit size is $\Theta(2^k)$ as well. For smaller ℓ, no matching total-size bound is implied by the product count alone.

□

D.2 A tropical disjunctive query

Over max–plus define

$$\mathcal{F}_k^{\max,+}(x, y) \coloneqq \max_z \sum_{i=1}^{k} \max(R_i(x, y), R_i(x, z)), \qquad S = \mathbb{R} \cup \{-\infty\}.$$

Proposition D.2 (Tropical disjunctive queries)**.** *For $k \geq 2$, the minimum product count and the minimum total circuit size for computing $\mathcal{F}_k^{\max,+}$ on all finite inputs are both $\Theta(2^k)$, even with arbitrary uniform entrywise functions and two-input max-aggregation kernels. The corresponding statement holds over min–plus after negating finite inputs and exchanging the two infinities.*

Proof. Use the full cube $\Lambda := 2^{[k]}$, and assign

$$R_i(\mu) := \begin{cases} 1/(|\mu|+1), & i \in \mu, \\ 0, & i \notin \mu, \end{cases} \qquad R_i(u,u) = -\infty.$$

The test colors are the colors λ with $|\lambda| \geq \lceil k/2 \rceil$, and $t_k^{\max,+} = \sum_{j=\lceil k/2 \rceil}^{k} \binom{k}{j} \geq 2^{k-1}$. Read at λ^c and use the fold of λ as before. Subtract the fixed read-edge baseline from a witness score. A color μ then contributes

$$\Delta_\lambda(\mu) = \frac{|\lambda \cap \mu|}{|\mu|+1} + |\lambda^c \cap \mu| \max\left\{ \frac{1}{|\mu|+1} - \frac{1}{|\lambda^c|+1}, 0 \right\}.$$

If $|\mu| < |\lambda|$, this is at most $|\mu|/(|\mu|+1)$, strictly below $|\lambda|/(|\lambda|+1)$. If $|\mu| \geq |\lambda| \geq |\lambda^c|$, the second term is zero; the first attains $|\lambda|/(|\lambda|+1)$ only for $\mu = \lambda$. Loop witnesses add zero to the baseline. Therefore λ is the unique maximizing color, and its fold strictly decreases the answer. The input uses only $O(\log k)$-bit rational amplitudes and $-\infty$.

The same theorem gives $\#_{\mathrm{mv}}(C) + 3\#_{\mathrm{mm}}(C) \geq t_k^{\max,+}$, including arbitrary max-aggregation kernels and arbitrary uniform entrywise functions. Distributing the maxima gives an upper bound of $2^k - 1$ row projections and $O(2^k)$ total nodes, since all subset sums can be shared. Thus total circuit size and product count are $\Theta(2^k)$. Randomization multiplies the lower count by $1 - 2\varepsilon$. Negating all finite scalar values and exchanging $-\infty$ with $+\infty$ gives the corresponding min–plus statement. These are exact-value lower bounds, not bounds for numerical approximation. □

E Directed targets and two-edge witness queries

We give a sharper bound without converse, then independently analyze the two-edge witness queries Ψ_k, Φ_k using anchors. All counting bounds below follow from Theorem 3.5, including its expected-cost extension with factor $1 - 2\varepsilon$.

E.1 A directed construction without converse

Corollary 4.6 (Without converse)**.** *Every converse-free relational circuit defining F_k on finite structures has at least B_k composition gates, for $k \geq 2$. Every such circuit defining G_k has at least 2^k composition gates, for $k \geq 1$. The latter bound is exact.*

Proof. Color each ordered pair of distinct vertices independently from a palette Λ; write $\chi(u,v)$ for its color, with no relation between $\chi(u,v)$ and $\chi(v,u)$. Require that for every u, v, now allowing $u = v$, and every $\mu, \nu \in \Lambda$, at least two vertices outside $\{u, v\}$ realize

$$(\chi(u,z), \chi(z,v)) = (\mu, \nu).$$

Candidate witnesses use disjoint ordered arcs, even when $u = v$. Lemma 3.1 and its deterministic construction therefore apply with the same threshold $n_0(N)$ and at most n^3N^2 bad events.

Reference arrays have a diagonal value $X[\mathbf{id}]$ and one value $X[\mu]$ per forward color. These tables are closed under entrywise operations and aggregation. Every output pair, including a diagonal, has outside-witness contribution $\kappa_* = \bigvee_{\mu,\nu} \kappa(X[\mu], Y[\nu])$; on a diagonal the sole endpoint contribution is $\kappa(X[\mathbf{id}], Y[\mathbf{id}])$. Recolor only the *outgoing* arcs at $\hbar$ by a map f, leaving all incoming arcs unchanged; predictions read the tables on this new forward coloring. Outside row $\hbar$, the product of the predicted operands equals the prediction of the product. On row $\hbar$, including $(\hbar, \hbar)$, the only altered outside-witness contribution is

$$\bigvee_{\mu \in f(\Lambda),\, \nu \in \Lambda} \kappa(X[\mu], Y[\nu]).$$

All endpoint contributions already have the predicted values. At most β first colors support this one join. Hence every aggregation notices at most β folds, and every Boolean composition at most one.

Use the same palettes, interpretations of the input relations, read edges, and folds as for F_k, G_k, with v_μ chosen so that the arc $(\hbar, v_\mu)$ has color μ. These queries inspect outgoing witness edges at $\hbar$, so their tests are unchanged. Theorem 3.5 gives

$$g_;(C) \geq B_k \quad (C \equiv_{\mathrm{fin}} F_k), \qquad g_;(C) \geq 2^k \quad (C \equiv_{\mathrm{fin}} G_k)$$

for converse-free relational circuits. The G_k bound is attained by its DNF on arbitrary inputs. The scalar bound also applies to the weighted F tests of Appendix D. Converse is excluded because a forward color table does not determine the reverse color; the coefficient-one argument consequently does not apply to full CoR.

□

E.2 The two-edge source formulas

The two-edge queries are

$$\Psi_k(x, y) \coloneqq \exists z \bigwedge_{i=1}^{k} (R_i(x, z) \vee R_i(z, y)) \qquad (k \geq 2),$$

$$\Phi_k(x, y) \coloneqq \exists z \bigwedge_{i=1}^{k} [(P_i(x, z) \wedge P_i(z, y)) \vee (Q_i(x, z) \wedge Q_i(z, y))] \qquad (k \geq 1).$$

The first is the source family used in [Nak22] for positive CoR. Their direct expansions on arbitrary inputs are

$$\Psi_k \equiv \bigcup_{I \subseteq [k]} (R_I; R_{I^c}), \qquad \Phi_k \equiv \bigcup_{A \subseteq [k]} (H_A; H_A).$$

Each uses 2^k compositions. Both witness edges pass through the quantified variable, unlike in F_k, G_k.

Theorem E.1 (Two-edge witness queries)**.** *Every full-*CoR *relational circuit defining Ψ_k on all finite structures has at least $\lceil B_k/2 \rceil$ composition gates, for $k \geq 2$. Every such circuit defining Φ_k has at least 2^{k-1} composition gates, for $k \geq 1$.*

The following construction and the calculation of predictions prove the theorem.

Take the symmetric reference core V with the middle-layer palette for Ψ_k, or the full cube for Φ_k. Add a disjoint set of vertices $\mathcal{A} \coloneqq \{\alpha_\lambda : \lambda \in \mathcal{H}\}$, called anchors. For Ψ_k, let every edge (α_λ, u), $u \in V$, satisfy R_i exactly when $i \in \lambda^c$, and interpret core edges as in §4.2. For Φ_k, let (α_λ, u) satisfy P_i exactly for $i \in \lambda$ and Q_i exactly for $i \in \lambda^c$, and interpret core edges as in §4.1. All core-to-anchor, anchor-to-anchor, and loop input entries are zero. For the test of λ, apply the fold of the core color λ to the edges at the hub, in both directions, leaving every anchor edge unchanged, and read the output at $(\alpha_\lambda, \hbar)$. A core witness satisfies the body exactly when its edge to $\hbar$ has color λ. The witness $\hbar$ fails because its loop inputs are zero and, for Ψ_k, because $(\alpha_\lambda, \hbar)$ misses R_i for $i \in \lambda \neq \varnothing$. Anchor witnesses α_μ also fail. The first witness edge $(\alpha_\lambda, \alpha_\mu)$ stays among the anchors and satisfies no input relation, which already decides Φ_k; for Ψ_k, moreover, $(\alpha_\mu, \hbar)$ misses R_i for every i in the nonempty set μ. Hence the reference answer is one and the folded answer is zero. There are B_k and 2^k tests, respectively.

E.3 Predictions in the presence of anchors

Every relation computed on the reference structure has one value on the core diagonal, one on each core color, one on each anchor-to-core and core-to-anchor class, and one on each ordered pair of anchors. Define predictions by recoloring the core entries and leaving all anchor classes fixed. These classes are closed under Boolean operations, converse, and composition: outside core witnesses supply all color pairs for distinct core endpoints, all equal-color pairs (μ, μ) on a core diagonal, all colors for an anchor/core pair, and a constant value for an anchor/anchor pair. Anchor witnesses contribute fixed values.

For a composition of X and Y, let S_X and S_Y be the sets of core colors on which they hold, and let J be the set of colors at the hub after the recoloring. It suffices that the three sets

$$S_X \cap J, \qquad S_Y \cap J, \qquad S_X \cap S_Y \cap J$$

are nonempty whenever S_X, S_Y, and $S_X \cap S_Y$ are. On core/core outputs this is precisely the local calculation of §3.4, with fixed anchor contributions added. On an output $(\alpha, \hbar)$, only a fixed anchor-to-core value conjoined with the second condition can change; on $(\hbar, \alpha)$, only the analogous first condition can change. For non-hub core endpoints the two-witness property leaves all colors available outside $\hbar$. On anchor/anchor outputs all predicted operands are unchanged. Thus the proof of Lemma 3.4 still applies, and each composition notices at most two folds. Read observations lie in unchanged anchor classes, so the observation condition (O) holds.

Applying Theorem 3.5 proves Theorem E.1. Since $|T| \geq 2\#_;(T) + 1$, the corresponding term-size lower bounds are $2\lceil B_k/2 \rceil + 1$ and $2^k + 1$. The composition bounds hold separately at every total size at least $n_0(N) + |\mathcal{H}|$, with the palette and target set used above. For an input-independent distribution with pointwise error at most $\varepsilon < 1/2$, the corresponding expected composition counts are at least $(1 - 2\varepsilon)B_k/2$ for Ψ_k and $(1 - 2\varepsilon)2^{k-1}$ for Φ_k. The input-occurrence refinement of §5.2 is specific to the unanchored construction.

The anchors provide the unchanged observations for these two-edge queries. On the same interpretations without anchors, the tests disappear. On the middle-layer recolored cores, $\Psi_k = \overline{\mathbf{id}}$. For Φ_k put $E \coloneqq \bigcap_{i=1}^{k}(P_i \cup Q_i)$; on the cube cores, $E = \overline{\mathbf{id}}$ and $\Phi_k = E; E = \mathbf{1}$. At distinct endpoints the surviving outside-witness color pairs supply complementary colors for Ψ_k and equal colors for

Φ_k; on a diagonal, symmetry prevents the former and permits the latter. Thus neither query detects a hub fold on these unanchored families.

E.4 A weighted two-edge query and its scope

Remark E.2 (Weighted two-edge queries). Over a nontrivial commutative idempotent semiring of breadth β, write the weighted two-edge query as

$$\Phi_k^{\mathrm{w}}(x,y) = \bigvee_{z\in\mathrm{dom}(\mathcal{M})} \bigotimes_{i=1}^{k} [(P_i(x,z)\otimes P_i(z,y)) \vee (Q_i(x,z)\otimes Q_i(z,y))].$$

The $0_S, 1_S$ anchor construction again gives 2^k tests. For ordinary semiring products, it suffices to retain the two core marginals $\bigvee_\mu X[\mu]$, $\bigvee_\mu Y[\mu]$ and the core diagonal join $\bigvee_\mu (X[\mu]\otimes Y[\mu])$. Distributivity reduces every anchor/core response to a fixed anchor coefficient multiplied by a marginal. At most 3β colors support these three joins, even if arbitrary uniform entrywise functions produce the intermediate operands. Consequently a circuit needs at least $2^k/(3\beta)$ semiring data-dimension products. Typed vector contractions obey the same bound, and inner products notice no fold, since reference vectors are constant on the core and predictions fix anchor coordinates. The expansion $\bigvee_A (H_A; H_A)$ gives the 2^k-product upper bound. The positive-semiring transfer of §6.4 also applies under its stated restrictions.

Arbitrary kernels cannot be substituted for multiplication in this last claim. On Boolean inputs, encode at each edge the set of all A for which H_A holds as one nonnegative real number. A two-input kernel returns one exactly when the two encoded sets intersect. One max-aggregation then computes Φ_k on every input. For Ψ_k, encode the set of true R_i and let the kernel test whether the union of its two decoded sets is $[k]$. This also uses one aggregation. Thus finite breadth alone is insufficient for the anchored two-edge queries with arbitrary kernels. The kernel lower bound for the read-edge/witness-edge query $\mathcal{G}_k$ in §6.3 has a different scope and remains valid.

F The classical single-exponential translation

We include the size analysis for the classical expressive equivalence [TG87, Mad06]. The relevant point is that successive quantifier eliminations need not create a tower of exponentials: an already translated subformula is treated as one leaf of the next Boolean expansion, without expanding its algebraic representation again.

Proposition F.1 (Classical translation). *Every* FO^3 *formula* φ *over binary relations with at most two free variables has an equivalent* CoR *term of size* $2^{O(|\varphi|)}$ *on all nonempty structures. Here size counts syntax-tree nodes.*

Proof. We represent a formula with at most two free variables as a binary relation of assignments to an ordered pair containing those variables. An unused coordinate is a dummy coordinate: membership is independent of it. Thus a unary formula is represented by a relation constant along rows or columns, and a sentence by $\mathbf{0}$ or $\mathbf{1}$. If a representation is moved to another ordered pair containing the same free variables, it is unchanged or replaced by its converse. This also handles the placement of a unary free variable in either coordinate. Closed formulas need no adjustment.

An atom with two distinct variables is represented by its input relation, its converse, or **id**, as appropriate. A reflexive atom $R(u,u)$, when u is the first coordinate, is represented by $(R \cap \mathbf{id}); \mathbf{1}$; when it is the second coordinate, use the converse. An equality $u = u$ is represented by $\mathbf{1}$. These translations have constant size. Boolean operations on formulas over a fixed pair become intersection, union, and complement.

Process quantified subformulas from the inside out. Consider $\exists w\, \psi$, and let u, v be the other two variable names, in that order. Its free variables are contained in $\{u, v\}$. In the body ψ, regard each atom and each maximal quantified subformula as one leaf of a Boolean formula, called its **shell**. Each leaf has at most two free variables: this is immediate for binary atoms, and a quantifier binds one of the three available names. The quantified leaves have already been translated. Crucially, the shell expansion treats each of their translations as an opaque term.

If the shell has r leaf occurrences, a truth-table DNF has at most 2^r clauses, each using at most r literals. Repeated or logically dependent leaves cause no problem; the propositional identity remains valid after substituting their actual interpretations. In each clause, assign its literals to three groups according to the free-variable pair: (u,v), (u,w), or (w,v). A literal with only one free variable is assigned to any group containing it; a closed literal can be put in the first group. Use converse to orient each translated literal for its chosen pair, and use complement for a negative literal. Writing the conjunctions of the groups as $A(u,v)$, $B(u,w)$, and $C(w,v)$, the clause translates as

$$\exists w\, (A(u,v) \wedge B(u,w) \wedge C(w,v)) \quad \longmapsto \quad A \cap (B; C).$$

An empty group is represented by $\mathbf{1}$. This identity also covers vacuous quantification because domains are nonempty. Take the union of the translated clauses. Universal quantification uses $\forall w\, \psi \equiv \neg \exists w\, \neg \psi$, which introduces no subformula duplication. At the root, any remaining Boolean shell with at most two free variables is translated directly, after orienting its leaves to a common ordered pair. This proves correctness on all nonempty structures by induction.

For the size bound, cut the original syntax tree at quantifiers into the shells just used. Count each child quantified subformula as one leaf of its parent's shell. The sum of the shell sizes, including these boundary leaves, is $O(|\varphi|)$: each original node is counted once, with only one extra boundary occurrence per quantified subformula. A shell of size s creates at most $2^{O(s)}$ local nodes and at most $2^{O(s)}$ copies of each child translation. Along any path from an original leaf or a newly introduced local node to the final output, the resulting duplication factor is therefore at most

$$\prod_{\text{shells on the path}} 2^{O(s)} \leq 2^{O(|\varphi|)}.$$

There are $O(|\varphi|)$ original nodes and shell locations, and the local exponential creation factor is included in this same bound. Summing their contributions gives $|\varphi|\, 2^{O(|\varphi|)} = 2^{O(|\varphi|)}$ nodes. No step expands the internal Boolean structure of a translated child, which is what prevents an iterated-exponential recurrence. □